2026 PRELIMS 99% MASTERY · CONSOLIDATED CHAPTER-WISE FACTS

CENSUS & URBANISATION

जनगणना एवं नगरीकरण — संपूर्ण उच्च-प्राथमिकता संकलन

Census 2011 Master Demographics, State Rankings, Literacy, Sex Ratio, Scheduled Castes/Tribes, Child Sex Ratio, Urbanisation Trends.

UPPCS & UKPCS Prelims 2026 • High-Yield Master Revision Series • 99% Coverage Edition

Census & Urbanisation — Master Chapter Index

Consolidated High-Yield Revision Matrix • All 7 Chapters

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2026 Prelims Apex Demographic Updates & Vital Statistics

YEAR / REF	KEY HIGH-YIELD FACT	EXAM RELEVANCE
2024–2026	**Digital Census Architecture & Self-Enumeration Portal:** Office of RGCCI preparing first fully digital, geo-referenced national census with mobile app enumeration across 16 scheduled languages.	Next Decennial Census Methodology
2023	**Bihar Caste-Based Survey (बिहार जाति गणना):** First published state headcount since 1931 showing EBC (36%), OBC (27.1%), SC (19.7%), ST (1.7%), General (15.5%).	Sub-categorization & Caste Demographics
2022–2023	**UN World Population Prospects (WPP 2022/2023):** Formally recognized India overtaking China as the world's most populous nation (~142.86 Crore in April 2023).	Global Demographic Rank Shifts
2021	**National Family Health Survey (NFHS-5):** National TFR declined to **2.0** (below replacement rate 2.1); UP TFR recorded at **2.35** (down from 2.74 in NFHS-4); Sex Ratio reached 1,020.	Fertility & Gender Ratio Benchmarks
2020	**Sample Registration System (SRS) Statistical Report:** National IMR declined to **28 per 1,000 live births**; UP IMR dropped to **38**; Under-5 Mortality (U5MR) at 32.	Vital Health Statistics Tracking
Rolling	**Census 2011 Statutory Baseline:** India Population: **121.08 Cr** (Decadal Growth 17.7%); UP Population: **19.98 Cr** (16.51% of India; 5th largest in world if considered a country).	Binding Official Reference Benchmark

Chapter 01: Census and Urbanisation Topic 1 — Census and Population Data

🎯 Consolidated Must-Score Facts

1. Under the Constitution of India, the **Census** is a subject enumerated in **Entry 69 of the Union List (List I)** of the Seventh Schedule.
2. The **Census Act, 1948** was piloted by **Sardar Vallabhbhai Patel** to provide permanent statutory authority for census operations.
3. Individual census information is strictly confidential and protected from judicial disclosure under **Section 15 of the Census Act, 1948**.
4. The **Registrar General and Census Commissioner of India (RGCCI)** operates under the **Ministry of Home Affairs (MHA)**.
5. The first non-synchronous census in modern India was conducted in **1872** during the viceroyalty of **Lord Mayo**.
6. The first complete synchronous decennial census of India was conducted in **1881** under **Lord Ripon**.
7. **W. C. Plowden** was the first Census Commissioner of India during the 1881 Census.
8. The **Year 1921** is known as the **"Year of the Great Divide"** in Indian demographic history because it recorded negative population growth (**-0.31%**).
9. The last caste-based census covering all castes in British India was conducted in **1931** under Census Commissioner **J. H. Hutton**.
10. The **1951 Census** was the **first census of independent India**, conducted under Census Commissioner **R. A. Gopalaswami**.
11. **Census 2011** was the **15th National Census** of India since 1872 and the **7th Census after Independence**.
12. The Census Commissioner of India for Census 2011 was **Dr. C. Chandramouli**.
13. The official motto of Census 2011 was **"Our Census, Our Future"**, with the mascot of a **Female Enumerator**.
14. The reference point for Census 2011 was **00:00 hours on 1st March 2011** (1st October 2010 for snowbound Himalayan areas).
15. India's total population as per Census 2011 stood at **1,210.85 Million (121.08 Crore)**.
16. India's population accounted for **17.5% of the total world population** in 2011, living on **2.4% of the world's land area**.
17. The **Decadal Population Growth Rate** of India during 2001–2011 was **17.70%** (Annual exponential rate of **1.64%**).
18. The population density of India in Census 2011 was **382 persons per square kilometer** (up from 325 in 2001).
19. The overall **Sex Ratio** of India in Census 2011 was **943 females per 1,000 males** (up from 933 in 2001).
20. The **Child Sex Ratio (0–6 years)** in Census 2011 dropped to an all-time low of **919 females per 1,000 males** (down from 927 in 2001).
21. The effective **Literacy Rate** of India in Census 2011 was **73.0%** (Male literacy: **80.9%**, Female literacy: **64.6%**).
22. The gender gap in literacy at the national level in 2011 stood at **16.3 percentage points**.
23. In Census 2011, **Kerala** had the highest literacy rate (**94.0%**), while **Bihar** had the lowest (**61.8%**).
24. **Kerala** had the highest sex ratio among Indian States (**1,084**), while **Haryana** had the lowest (**879**).
25. Among Union Territories, **Puducherry** had the highest sex ratio (**1,037**), while **Daman and Diu** had the lowest (**618**).
26. **Meghalaya** recorded the highest decadal population growth rate among States (**27.95%**), followed by Arunachal Pradesh (26.03%).
27. **Nagaland** was the only Indian State to record a **negative decadal growth rate (-0.58%)** during 2001–2011.
28. **Bihar** has the highest population density among States (**1,106 persons / sq km**), followed by West Bengal (1,028).
29. **Arunachal Pradesh** has the lowest population density in India (**17 persons / sq km**).
30. Among UTs, **Delhi** has the highest density (**11,320 persons / sq km**), while **Andaman & Nicobar Islands** has the lowest (**46**).
31. The Scheduled Caste (SC) population in India was **20.14 Crore (16.63% of total population)** in 2011.
32. The Scheduled Tribe (ST) population in India was **10.43 Crore (8.61% of total population)** in 2011.
33. **Punjab** has the highest proportion of Scheduled Castes (**31.9%**), while **Uttar Pradesh** has the largest absolute SC population.
34. **Madhya Pradesh** has the highest absolute Scheduled Tribe population, while **Lakshadweep (94.8%)** and **Mizoram (94.4%)** have the highest ST percentages.
35. The States of **Punjab, Haryana, Delhi, Chandigarh, and Puducherry** have **NO notified Scheduled Tribe population**.
36. The States of **Nagaland, Mizoram, Meghalaya, Lakshadweep, and Andaman & Nicobar Islands** have **NO notified Scheduled Caste population**.
37. The proportion of **Urban Population** in India as per Census 2011 was **31.16%** (Rural: **68.84%**).
38. **Goa** is the most urbanised State in India (**62.2% urban population**), while **Himachal Pradesh** is the least urbanised (**10.0%**).
39. The total population of **Uttar Pradesh** in Census 2011 was **19.98 Crore (199,812,341)**, accounting for **16.51% of India's population**.
40. The decadal growth rate of Uttar Pradesh during 2001–2011 was **20.22%**.
41. The population density of Uttar Pradesh in 2011 was **829 persons per square kilometer** (4th highest among States).
42. The sex ratio of Uttar Pradesh in 2011 was **912** (Child Sex Ratio: **902**).
43. The overall literacy rate of Uttar Pradesh in 2011 was **67.7%** (Male: **77.3%**, Female: **57.2%**; Gap: **20.1%**).

44. The most populous district of Uttar Pradesh is **Prayagraj (Allahabad)** (59.54 Lakh), and the least populous is **Mahoba** (8.75 Lakh).
45. **Gautam Buddha Nagar** recorded the highest decadal growth rate in UP (**49.1%**), while **Kanpur Nagar** recorded the lowest (**9.9%**).
46. **Ghaziabad** has the highest population density in UP (**3,971 / km^2**), while **Lalitpur** has the lowest (**242 / km^2**).
47. **Jaunpur** has the highest sex ratio in UP (**1,024**), while **Gautam Buddha Nagar** has the lowest (**851**).
48. **Gautam Buddha Nagar** has the highest literacy rate in UP (**80.12%**), while **Shravasti** has the lowest (**46.74%**).
49. **Kanpur Nagar** has the highest female literacy rate in UP (**75.05%**), while **Shravasti** has the lowest female literacy (**34.78%**).
50. The **Sample Registration System (SRS)** under the Registrar General of India is the premier annual source for estimating IMR, MMR, CBR, and CDR in India.

Timeline / Chronology

Chronological Sequence

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1872 ⟶ First Non-Synchronous Census (Lord Mayo)
1881 ⟶ First Synchronous Decennial Census (Lord Ripon; W.C. Plowden, 1st Commissioner)
1921 ⟶ "Year of the Great Divide" (Negative Growth: -0.31%, Famine/Flu Pandemic)
1931 ⟶ Last Caste Census under British India (J.H. Hutton)
1948 ⟶ The Census Act, 1948 enacted (Sardar Vallabhbhai Patel)
1951 ⟶ First Census of Independent India (R.A. Gopalaswami; 1951 Census Act applied)
2001 ⟶ 14th National Census (First of the 21st Century)
2011 ⟶ 15th National Census / 7th Post-Independence (Dr. C. Chandramouli; 121.08 Cr)
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Table 2: Historical Timeline of Indian Census (1872 to Present)

CENSUS YEAR / MILESTONE	VICEROY / BRITISH RULE / LEADERSHIP	CENSUS COMMISSIONER / REGISTRAR GENERAL	OFFICIAL MOTTO / HISTORICAL SIGNIFICANCE	KEY FACTS & DEMOGRAPHIC MILESTONES
1872	**Lord Mayo** (1869–1872)	Non-centralized, local supervision	**First Census in Modern India** (Non-synchronous)	Conducted asynchronously across various provinces between 1867 and 1872. Did not cover all British territories.
1881	**Lord Ripon** (1880–1884)	**W. C. Plowden** (First Census Commissioner)	**First Synchronous Decennial Census**	First complete, uniform, decennial enumeration covering the entire territory of British India and Princely States.
1891 to 1911	Decennial Censuses under British Raj	J. A. Baines (1891), H. H. Risley (1901), E. A. Gait (1911)	Caste and ethnographic classifications	Risley (1901) classified Indian society by social precedence and anthropometric measurements.
1921	Lord Reading	J. T. Marten	**"The Year of the Great Divide"**	Only Census in Indian history with **Negative Population Growth (-0.31%)** due to 1918 Spanish Influenza pandemic, plague, and famines.
1931	Lord Irwin / Willingdon	**J. H. Hutton**	**Last Caste Census in British India**	Last enumeration to record complete caste and sub-caste data for all Indians (basis of Mandal Commission reservations in 1980/1990).
1941	Lord Linlithgow	M. W. M. Yeatts	Restricted wartime census	Conducted during World War II; detailed tabulation tables were not published due to wartime paper shortages.
1948	Post-Independence Government (Sardar Patel)	Drafted by M. W. M. Yeatts	**Enactment of the Census Act, 1948**	Provided permanent statutory authority for decennial census operations without requiring ad-hoc legislation.
1951	Independent India (Jawaharlal Nehru)	**R. A. Gopalaswami**	**1st Census of Independent India** (8th Modern Census)	Introduced National Register of Citizens (NRC 1951); documented partition refugee resettlement and land ownership.
1971	Indira Gandhi Government	A. Chandra Sekhar	Centenary Census (1872–1971)	Recorded peak decadal population growth (**24.80%**); introduced economic activity classification into main/marginal workers.
2011	UPA-II Government	**Dr. C. Chandramouli**	"*Our Census, Our Future*" Mascot: **Female Enumerator**	**15th National Census** since 1872, and **7th Census after Independence**. Covered 640 districts, 5,924 sub-districts, and 7,935 towns.

📊 Quick Reference Grid

Table 1: Census 2011 Master Demographic Matrix (India vs Uttar Pradesh vs Uttarakhand)

DEMOGRAPHIC PARAMETER / INDICATOR	ALL-INDIA (CENSUS 2011)	UTTAR PRADESH (CENSUS 2011)	UTTARAKHAND (CENSUS 2011)	EXAM HIGH-YIELD NOTES & DISTINCTIONS
Total Population	**1,210.85 Million (121.08 Cr)**	**199.81 Million (19.98 Cr)**	**10.09 Million (1.01 Cr)**	• UP is India's **most populous State** (16.51% of national total). • If UP were a country, it would be the 5th most populous globally (after China, India, US, Indonesia).
Male : Female Share (%)	Males: **51.54%** (62.31 Cr) Females: **48.46%** (58.74 Cr)	Males: **52.29%** (10.44 Cr) Females: **47.71%** (9.53 Cr)	Males: **50.93%** (5.14 Cr) Females: **49.07%** (4.95 Cr)	Consistent slight male preponderance across all three demographic units.
Decadal Growth Rate (2001–2011)	**17.70%** (Rural: 12.30%, Urban: 31.80%)	**20.22%** (Rural: 17.99%, Urban: 28.82%)	**18.81%** (Rural: 11.52%, Urban: 39.95%)	• National decadal growth declined from 21.54% (1991–2001). • UP growth (20.22%) is higher than national average. • UK hill districts (Almora, Pauri) registered **negative growth**.
Population Density (per sq km)	**382 persons / km²** (up from 325 in 2001)	**829 persons / km²** (4th highest state after Bihar, WB, Kerala)	**189 persons / km²** (Mountainous terrain)	• Bihar has highest density among States (**1,106**). • Delhi has highest density among UTs (**11,320**). • Arunachal Pradesh has lowest density (**17**).
Overall Sex Ratio (Females per 1000 Males)	**943** (up from 933 in 2001)	**912** (Lower than national average)	**963** (Substantially higher than national average)	• Kerala highest among States (**1,084**); Daman & Diu lowest UT (**618**). • In UP, **Jaunpur** has highest sex ratio (**1,024**).
Child Sex Ratio (0–6 Years)	**919** (Down from 927 in 2001 — Alarming Decline)	**902** (Down from 916 in 2001)	**890** (Down from 908 in 2001)	• Mizoram highest (**970**); Haryana lowest (**834**). • Child sex ratio reflects entrenched female foeticide and daughter-deficit.
Effective Literacy Rate (7+ Years)	**73.0%** (Males: 80.9%, Females: 64.6%)	**67.7%** (Males: 77.3%, Females: 57.2%)	**78.8%** (Males: 87.4%, Females: 70.0%)	• Kerala highest (**94.0%**); Bihar lowest (**61.8%**). • Male-Female literacy gap in India: **16.3%**; in UP: **20.1%**; in UK: **17.4%**.
Scheduled Caste (SC) Population	**20.14 Crore (16.63%)**	**4.14 Crore (20.70%)** — Largest SC population in India	**18.93 Lakh (18.76%)**	• UP has highest *absolute number* of SCs. • Punjab has highest *percentage* of SCs (**31.9%**).
Scheduled Tribe (ST) Population	**10.43 Crore (8.61%)**	**11.34 Lakh (0.57%)**	**2.92 Lakh (2.89%)**	• MP has highest *absolute number* of STs (1.53 Cr). • Lakshadweep (94.8%) & Mizoram (94.4%) have highest *percentage* of STs.
Urban : Rural Proportion (%)	Urban: **31.16%** Rural: **68.84%**	Urban: **22.27%** Rural: **77.73%**	Urban: **30.23%** Rural: **69.77%**	• Goa is most urbanised state (**62.2%**); Himachal Pradesh is least urbanised (**10.0%**). • UP has 2nd largest absolute urban population after Maharashtra.

Table 3: Uttar Pradesh District-Level Demographic Extremes (Census 2011)

DEMOGRAPHIC INDICATOR	TOP RANKING DISTRICTS IN UTTAR PRADESH (HIGHEST)	BOTTOM RANKING DISTRICTS IN UTTAR PRADESH (LOWEST)
Total Population	**1. Prayagraj (Allahabad)** (59.54 Lakh) 2. Moradabad (47.72 Lakh) 3. Ghaziabad (46.81 Lakh) 4. Azamgarh (46.13 Lakh) 5. Lucknow (45.89 Lakh)	**1. Mahoba** (8.75 Lakh) 2. Chitrakoot (9.91 Lakh) 3. Hamirpur (11.04 Lakh) 4. Shravasti (11.17 Lakh) 5. Lalitpur (12.21 Lakh)
Decadal Population Growth Rate (%)	**1. Gautam Buddha Nagar (49.1%)** 2. Ghaziabad (42.3%) 3. Shravasti (30.5%) 4. Bahraich (29.3%) 5. Balrampur (27.7%)	**1. Kanpur Nagar (9.9%)** 2. Hamirpur (11.1%) 3. Baghpat (11.9%) 4. Fatehpur (14.1%) 5. Deoria (14.2%)
Population Density (persons / km²)	**1. Ghaziabad (3,971)** 2. Varanasi (2,395) 3. Lucknow (1,816) 4. Bhadohi / Sant Ravidas Nagar (1,555) 5. Kanpur Nagar (1,452)	**1. Lalitpur (242)** 2. Sonbhadra (270) 3. Hamirpur (275) 4. Mahoba (279) 5. Chitrakoot (308)
Sex Ratio (Females per 1000 Males)	**1. Jaunpur (1,024)** 2. Azamgarh (1,019) 3. Deoria (1,017) 4. Pratapgarh (998) 5. Sultanpur (983)	**1. Gautam Buddha Nagar (851)** 2. Hamirpur & Kanpur Nagar (861 each) 3. Banda (863) 4. Mathura (863) 5. Auraiya (864)
Child Sex Ratio (0–6 Years)	**1. Balrampur (950)** 2. Sant Kabir Nagar (942) 3. Bahraich (935) 4. Siddharthnagar (935) 5. Barabanki (932)	**1. Baghpat (841)** 2. Gautam Buddha Nagar (843) 3. Ghaziabad (850) 4. Meerut (852) 5. Bulandshahr (854)
Overall Literacy Rate (%)	**1. Gautam Buddha Nagar (80.12%)** 2. Kanpur Nagar (79.65%) 3. Auraiya (78.95%) 4. Etawah (78.41%) 5. Ghaziabad (78.07%)	**1. Shravasti (46.74%)** 2. Bahraich (49.36%) 3. Balrampur (49.51%) 4. Badaun (51.29%) 5. Rampur (53.34%)
Female Literacy Rate (%)	**1. Kanpur Nagar (75.05%)** 2. Lucknow (71.54%) 3. Gautam Buddha Nagar (70.82%) 4. Auraiya (70.41%) 5. Ghaziabad (69.79%)	**1. Shravasti (34.78%)** 2. Balrampur (38.43%) 3. Bahraich (39.18%) 4. Badaun (40.09%) 5. Rampur (44.44%)

🧠 Memory Anchors & Mnemonics

- **State Literacy Top 4**: **K-M-G-T** (Kerala 94% > Mizoram 91.3% > Goa 88.7% > Tripura 87.2%)
- **UP Density Extremes**: Highest = **G-V-L** (Ghaziabad 3,971 > Varanasi 2,395 > Lucknow 1,816) | Lowest = **L-S-H** (Lalitpur 242 < Sonbhadra 270 < Hamirpur 275)
- **UP Sex Ratio Extremes**: Highest = **J-A-D** (Jaunpur 1,024 > Azamgarh 1,019 > Deoria 1,017) | Lowest = **GB-B/H** (Gautam Buddha Nagar 851 < Baghpat/Hamirpur 861)
- **Zero Scheduled Tribes**: **PH-DC-P** (Punjab, Haryana, Delhi, Chandigarh, Puducherry have NO STs)
- **Zero Scheduled Castes**: **N-M-M-L-A** (Nagaland, Mizoram, Meghalaya, Lakshadweep, A&N have NO SCs)
- **Modified Counting Approach**: India follows **Extended De Facto / Modified De Jure** (enumerated at habitual residence over 20-day window, homeless on final night).

Chapter 02: Census and Urbanisation Topic 2 — Population Growth, Demographic Transition and Theories

🎯 Consolidated Must-Score Facts

1. **Thomas Robert Malthus** published his seminal work *An Essay on the Principle of Population* in **1798**.
2. Malthus postulated that population increases in **Geometric Progression** ($1, 2, 4, 8, 16...$) while food production increases in **Arithmetic Progression** ($1, 2, 3, 4, 5...$).
3. According to Malthus, unchecked population has the biological tendency to **double every 25 years**.
4. Malthusian checks are classified into two categories: **Preventive Checks** (man-made, moral restraint) and **Positive Checks** (nature-induced, famines/wars/diseases).
5. The **Demographic Transition Theory (DTT)** was originally formulated by **Warren S. Thompson (1929)** and developed by **Frank W. Notestein (1945)**.
6. The Demographic Transition Theory explains the transition from high birth and death rates to low birth and death rates as an economy modernizes.
7. **Stage 1 (High Stationary)** of DTT is characterized by high birth rates, high death rates, and a stable, low population.
8. **Stage 2 (Early Expanding)** of DTT is characterized by high birth rates and rapidly falling death rates, leading to a **Population Explosion**.
9. **Stage 3 (Late Expanding)** of DTT is characterized by rapidly falling birth rates and low death rates, with population growth slowing down.
10. **Stage 4 (Low Stationary)** of DTT is characterized by low birth rates, low death rates, and a stable, mature population.
11. **India is currently in Stage 3 (Late Expanding Phase)** of the Demographic Transition.
12. **India prior to 1921** was in **Stage 1 (High Stationary Phase)** of the Demographic Transition.
13. The **Optimum Population Theory** was propounded by English economist **Edwin Cannan** in his 1924 book *Wealth*.
14. Optimum population is defined as that population size which yields the **maximum output / real income per capita**.
15. **Hugh Dalton** formulated the degree of maladjustment equation: **$M = (A - O) / O$**, where A is Actual Population and O is Optimum Population.
16. If $M > 0$ in Dalton's formula, the country is suffering from **Overpopulation**; if $M < 0$, it is **Underpopulated**.
17. **Karl Marx** rejected Malthusian population theory, arguing that overpopulation and poverty are products of **capitalist exploitation**, not natural laws.
18. Marx introduced the concept of the **"Industrial Reserve Army of Labour"**, created by machinery displacing human workers.
19. **Ester Boserup (1965)** propounded that population pressure acts as an engine of agricultural innovation and intensification.
20. In Indian demographic history, **Phase I (1901–1921)** was the period of **stagnant / stationary population**.
21. The **Year 1921** is universally termed **"The Year of the Great Divide"** because population registered a negative growth of **-0.31%**.
22. The 1918 **Influenza (Spanish Flu) Pandemic** killed an estimated 14 million people (approx. 5% of India's population), causing negative growth in 1911–1921.
23. **Phase II (1921–1951)** in India was characterized by **steady, moderate population growth**.
24. **Phase III (1951–1981)** in India was characterized by **rapid high growth / Population Explosion**.
25. India's decadal population growth rate peaked historically during the decade **1961–1971 at 24.80%**.
26. **Phase IV (1981–2011+)** in India is characterized by **high growth with clear signs of slowing down**.
27. India's decadal population growth rate dropped from **21.54%** (1991–2001) to **17.70%** (2001–2011).
28. India's annual exponential population growth rate in Census 2011 was **1.64%** (down from 1.97% in 2001).
29. **Natural Increase of Population** is calculated as: **Crude Birth Rate (CBR) minus Crude Death Rate (CDR)**.
30. According to the **Rule of 70**, population doubling time is approximately **70 divided by the annual percentage growth rate**.
31. At an annual growth rate of 1.64%, India's population doubling time is approximately **42.7 years**.
32. The concept of **Demographic Dividend** refers to the economic growth advantage when the share of the working-age population (**15–59 years**) exceeds the dependent population.
33. India entered its demographic dividend window around **2005–06**, which is expected to last until roughly **2055–56**.
34. The **Dependency Ratio** is calculated as: $[(ext{Population } 0-14 + ext{Population } 60+) / ext{Working-Age Population } 15-59] imes 100$.
35. The **Total Fertility Rate (TFR)** required for population stabilization (Replacement Level) is **2.1 children per woman**.
36. As per the **National Family Health Survey-5 (NFHS-5, 2019-21)**, India's national TFR declined to **2.0**, dropping below the replacement level of 2.1.
37. **Henry George** in *Progress and Poverty* (1879) argued that poverty arises from land monopoly and landlord rent extraction, not overpopulation.

38. **Paul R. Ehrlich** published the Neo-Malthusian bestseller *The Population Bomb* in **1968**.
39. The **Club of Rome** published the influential environmental report *The Limits to Growth* in **1972**, warning of resource exhaustion from exponential growth.
40. The **Simon-Ehrlich Wager (1980)** between Julian Simon (cornucopian economist) and Paul Ehrlich proved Ehrlich wrong when prices of 5 raw metals fell over 10 years.
41. **Julian Simon** in *The Ultimate Resource* (1981) argued that human intellect and labour are the ultimate economic resource.
42. A **Population Pyramid** with a broad base and sharply tapering top indicates a **young population with high birth and high mortality rates** (Stage 1 or 2).
43. A **Stationary / Bell-shaped Population Pyramid** with nearly equal base and middle indicates **low birth and death rates** (Stage 3 or 4).
44. An **Inverted / Urn-shaped Population Pyramid** with a narrow base and broad top indicates an **aging population with sub-replacement fertility** (Stage 5).
45. In India, **Kerala and Tamil Nadu** were the first major States to achieve below-replacement level fertility (TFR < 2.1).
46. Among large Indian States, **Bihar (TFR ~3.0)** and **Uttar Pradesh (TFR ~2.4 in NFHS-5)** still retain fertility above replacement level.
47. India was the **first country in the world** to launch a National Family Planning Programme in **1952**.
48. The **National Population Policy (NPP) 2000** aimed to achieve population stabilization in India by the year **2045** (subsequently revised to 2070).
49. The **J-Curve of Population Growth** represents exponential / unrestricted population growth under ideal environmental conditions.
50. The **S-Curve (Logistic Growth Curve)** represents population growth that stabilizes at the environment's **Carrying Capacity (K)**.

⏱ Timeline / Chronology

Chronological Sequence

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Phase I   (1901–1921) ⟶ Period of Stagnant / Stationary Growth (High BR + High DR; 1921 Divide: -0.31%)
Phase II  (1921–1951) ⟶ Period of Steady / Controlled Growth (Mortality decline begins; Sanitation/Health)
Phase III (1951–1981) ⟶ Period of Rapid High Growth / "Population Explosion" (High BR + Sharp DR drop)
Phase IV  (1981–2011) ⟶ Period of High Growth with Definite Signs of Slowing Down (Fertility decline begins)
Post-2011 (Present)   ⟶ Declining Fertility toward Sub-Replacement (NFHS-5: TFR 2.0 nationally)
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Table 1: Four Distinct Phases of India's Demographic Growth (1901 to Present)

DEMOGRAPHIC PHASE	PERIOD / YEARS	BIRTH RATE & DEATH RATE BEHAVIOR	DECADAL GROWTH & CHARACTERISTICS	UNDERLYING SOCIO-ECONOMIC & HEALTH DRIVERS
Phase I: Period of Stagnant / Stationary Population	**1901–1921**	• **High Birth Rate** (~48.1 per 1000) • **High Death Rate** (~47.2 per 1000)	Net growth was negligible (**5.4%** in 1901–11, **-0.31%** in 1911–21). Population increased by only 13 million over 20 years.	• Severe recurring famines (1896, 1899). • Epidemics of plague, cholera, malaria. • **1918 Influenza (Spanish Flu) pandemic** killed ~14 million Indians (~5% of population). • Widespread malnutrition and absence of modern healthcare.
Phase II: Period of Steady / Moderate Growth	**1921–1951**	• **High Birth Rate** (~40 per 1000) • **Declining Death Rate** (~27 per 1000)	Moderate, steady expansion (**11.0%** in 1921–31, **14.2%** in 1931–41, **13.3%** in 1941–51). Total population grew by 110 million.	• Famine prevention and better transport/grain distribution. • Improved sanitation and public health measures. • Introduction of vaccines and sulpha drugs curbing major epidemics. • **1921 is "The Year of the Great Divide"**.
Phase III: Period of Rapid High Growth (Population Explosion)	**1951–1981**	• **High / Slow-declining Birth Rate** (~37 per 1000) • **Rapidly Plummeting Death Rate** (~15 per 1000)	**Population Explosion**: Decadal growth surged to **21.5%** (1951–61), **24.8%** (1961–71 — historical peak), and **24.7%** (1971–81). Population nearly doubled from 361 million to 683 million.	• Massive post-independence health interventions (eradication of smallpox, control of malaria with DDT). • Introduction of broad-spectrum antibiotics (Penicillin). • Expansion of hospital infrastructure and maternal care. • Traditional pro-natalist social attitudes and early universal marriage persisted.
Phase IV: Period of High Growth with Definite Signs of Slowing Down	**1981–2011+**	• **Rapidly Declining Birth Rate** (from 34 to ~20 per 1000) • **Low Stable Death Rate** (~7 per 1000)	Decadal growth decelerated to **23.9%** (1981–91), **21.5%** (1991–2001), and **17.7%** (2001–2011). Annual exponential growth rate dropped to **1.64%**.	• Rise in female literacy and female age at marriage. • Expansion of family planning and modern contraception. • Urbanisation, nuclear family preferences, and rising cost of child rearing. • TFR reached **2.0** (below replacement level 2.1 in NFHS-5).

📊 Quick Reference Grid

Table 2: Comparative Matrix of Classical & Modern Demographic Theories

DEMOGRAPHIC THEORY	PROPOUNDER / KEY SCHOLARS	PUBLICATION & YEAR	CENTRAL HYPOTHESIS & CORE DOCTRINE	PRIMARY MECHANISMS & SOLUTIONS
Malthusian Theory of Population	**Thomas Robert Malthus** (English Economist & Clergyman)	*An Essay on the Principle of Population* (**1798**)	• Population increases in **Geometric Progression** ($1, 2, 4, 8, 16, 32...$), doubling every 25 years. • Food production increases in **Arithmetic Progression** ($1, 2, 3, 4, 5, 6...$) due to diminishing returns to land. • Population growth inevitably outstrips food supply, causing widespread starvation (**Malthusian Trap**).	Two checks to restore balance: 1. **Preventive Checks** (Man-made): Moral restraint, celibacy, delayed marriage. 2. **Positive Checks** (Nature's cruel intervention): Famines, pestilence, epidemics, wars.
Demographic Transition Theory (DTT)	**Warren S. Thompson** (1929), **Frank W. Notestein** (1945), **C. P. Blacker** (1947)	*Plenty of People* (Notestein, 1945)	Population growth passes through sequential stages driven by economic development and industrialization: • Stage 1: High Birth & Death Rates $\implies$ Low Growth. • Stage 2: High Birth & Falling Death Rates $\implies$ Population Explosion. • Stage 3: Falling Birth & Low Death Rates $\implies$ Slowing Growth. • Stage 4: Low Birth & Low Death Rates $\implies$ Stationary Population.	Transition occurs naturally as agrarian traditional societies industrialise, urbanise, educate women, and adopt modern healthcare.
Optimum Population Theory	**Edwin Cannan** (1924); refined by **Alexander Carr-Saunders**, **Hugh Dalton**, **Lionel Robbins**	*Wealth* (Edwin Cannan, 1924)	**Optimum Population** is that specific size of population which, given natural resources, capital stock, and state of technology, yields the **Maximum Per Capita Output / Maximum Real Income per Head**.	**Dalton's Maladjustment Formula**: $$M = ↑rac{A - O}{O}$$ • $M > 0$: Overpopulation. • $M < 0$: Underpopulation. • $M = 0$: Perfect Optimum.
Marxian Theory of Population	**Karl Marx** (German Philosopher & Economist)	*Das Kapital* (**1867**)	Rejected Malthus completely. Held that **there is no universal, eternal law of population**. Overpopulation and poverty are inherent structural evils created by the **capitalist mode of production** and unequal distribution of wealth.	Capitalists accumulate machinery, displacing workers to create an **"Industrial Reserve Army of Labour"** to depress wages. Solution: Socialist restructuring and equitable distribution.
Boserupian Agricultural Intensification Hypothesis	**Ester Boserup** (Danish Economist)	*The Conditions of Agricultural Growth* (**1965**)	**Population pressure acts as a stimulus** to technological innovation and agricultural intensification ("Necessity is the mother of invention"), reversing Malthusian fatalism.	Population growth forces shift from fallowing/shifting cultivation to multi-cropping, irrigation, fertilizers, and modern mechanization.

Table 3: Detailed Stages of Demographic Transition Theory (DTT)

STAGE OF TRANSITION	STAGE NAME & ECONOMIC CHARACTER	BIRTH RATE (CBR)	DEATH RATE (CDR)	NATURAL GROWTH RATE	AGE STRUCTURE & POPULATION PYRAMID	GLOBAL & INDIAN EXAMPLES
Stage 1	**High Stationary** (Pre-Industrial / Agrarian)	**Very High** (35–45 per 1000)	**Very High & Fluctuating** (30–40 per 1000)	**Negligible / Zero** ($\approx 0\%$)	High infant mortality; concave wide-base, very narrow-top pyramid (low life expectancy ~30 years).	Remote indigenous tribes; Britain before 1750; **India prior to 1921**.
Stage 2	**Early Expanding (Population Explosion)**	**Very High** (30–40 per 1000)	**Plummeting Rapidly** (10–20 per 1000)	**Extremely High** (2.0% to 3.5% annually)	Broad-base, expanding triangular pyramid; high child dependency ratio.	Sub-Saharan African countries (Niger, Chad, Mali); **India during 1951–1981**.
Stage 3	**Late Expanding (Slowing Growth)**	**Declining Rapidly** (15–25 per 1000)	**Low & Stabilized** (6–10 per 1000)	**Moderate to Low** (1.0% to 1.5%)	Bulging middle age group; demographic dividend phase; bell-shaped pyramid.	**India currently (Census 2011 to present)**, Brazil, South Africa, Mexico.
Stage 4	**Low Stationary (Stabilized)**	**Low** (10–14 per 1000)	**Low** (8–12 per 1000)	**Near Zero / Stable** ($\le 0.5\%$)	Barrel / urn-shaped pyramid; high median age; aging population.	United States, Canada, United Kingdom, Australia, China.
Stage 5	**Declining / Contracting (Negative Growth)**	**Extremely Low** (below CDR, < 9 per 1000)	**Rising slightly due to old age** (10–12 per 1000)	**Negative Natural Growth** ($< 0\%$)	Inverted pyramid; shrinking youth population; acute elderly dependency.	Japan, Germany, Italy, South Korea, Russia.

🧠 Memory Anchors & Mnemonics

- **DTT 4 Stages**: **H-E-L-L** (High Stationary -> Early Expanding -> Late Expanding -> Low Stationary)
- **Malthusian Checks**: **P-P** = **Preventive** (celibacy, late marriage, moral restraint) vs **Positive** (famine, war, pestilence, natural misery)
- **Arithmetic vs Geometric Growth**: Food grows **Arithmetically** (1, 2, 3, 4, 5...) while Population grows **Geometrically** (1, 2, 4, 8, 16...)
- **Optimum Population Formula**: Dalton's Maladjustment $M = (A - O) / O$ (If $A > O \implies$ Overpopulation; $A < O \implies$ Underpopulation)
- **Decadal Growth Peak**: Peak decadal growth in India was recorded in **1961–1971 (24.80%)**.

Chapter 03: Census and Urbanisation Topic 3 — Population Composition and Demographic Characteristics

🎯 Consolidated Must-Score Facts

1. The proportion of the **Working-Age Population (15–59 years)** in India as per Census 2011 was **60.3%**.
2. The proportion of the **Child Population (0–14 years)** in India in 2011 was **30.8%**, while the **Senior Citizen Population (60+ years)** was **8.6%**.
3. India's **Total Dependency Ratio** in Census 2011 was **65.2%** (down from 75.2% in 2001).
4. In Census classification, a **Main Worker** is an individual who engaged in economic activity for **183 days (6 months) or more** in the preceding year.
5. A **Marginal Worker** is an individual who worked for **less than 183 days** during the preceding reference year.
6. Among India's workforce in 2011, **Agricultural Labourers (30.0%)** formed the largest single occupational category, surpassing Cultivators (24.6%).
7. The overall **Sex Ratio** of India in Census 2011 was **943 females per 1,000 males**.
8. **Kerala** has the highest sex ratio among Indian States at **1,084**, followed by Tamil Nadu (996) and Andhra Pradesh (993).
9. **Haryana** has the lowest sex ratio among Indian States at **879**, followed by Jammu & Kashmir (889) and Sikkim (890).
10. **Puducherry** has the highest sex ratio among Union Territories at **1,037**, while **Daman & Diu** has the lowest at **618**.
11. The **Child Sex Ratio (0–6 years)** of India in Census 2011 was **919**, dropping from 927 in 2001.
12. **Mizoram** recorded the highest Child Sex Ratio among States at **970**, followed by Meghalaya (970) and Chhattisgarh (969).
13. **Haryana** recorded the lowest Child Sex Ratio among States at **834**, followed by Punjab (846).
14. Effective Literacy Rate in India is calculated strictly for the population aged **7 years and above**.
15. India's effective literacy rate in Census 2011 was **73.0%** (Male: **80.9%**, Female: **64.6%**).
16. The national **Gender Gap in Literacy** in Census 2011 was **16.3 percentage points**.
17. **Kerala** has the highest overall literacy rate (**94.0%**) as well as highest female literacy (**92.1%**).
18. **Bihar** has the lowest overall literacy rate (**61.8%**) and the lowest male literacy rate (**71.2%**).
19. **Rajasthan** recorded the lowest female literacy rate among all States at **52.1%** (and the widest gender literacy gap of **27.1%**).
20. **Meghalaya** recorded the narrowest gender gap in literacy in India at only **3.1 percentage points**.
21. The **Scheduled Caste (SC)** population in India in 2011 was **20.14 Crore**, constituting **16.63%** of the national total.
22. The **Scheduled Tribe (ST)** population in India in 2011 was **10.43 Crore**, constituting **8.61%** of the national total.
23. The Sex Ratio among **Scheduled Castes** in India is **945**, which is higher than the national average (943).
24. The Sex Ratio among **Scheduled Tribes** in India is **990**, which is the highest among all major social groups.
25. The Child Sex Ratio (0–6 years) among **Scheduled Tribes** is **957** (compared to 919 at all-India level).
26. **Uttar Pradesh** has the largest absolute Scheduled Caste population in India (**4.14 Crore**).
27. **Punjab** has the highest proportion of Scheduled Castes relative to its total population (**31.9%**).
28. **Madhya Pradesh** has the largest absolute Scheduled Tribe population in India (**1.53 Crore**).
29. **Lakshadweep (94.8%)** and **Mizoram (94.4%)** have the highest percentage of Scheduled Tribes in their total population.
30. The States/UTs of **Punjab, Haryana, Chandigarh, Delhi, and Puducherry** have **NO notified Scheduled Tribe population**.
31. The States/UTs of **Nagaland, Mizoram, Meghalaya, Lakshadweep, and Andaman & Nicobar Islands** have **NO notified Scheduled Caste population**.
32. In India, there are **75 Particularly Vulnerable Tribal Groups (PVTGs)** across 18 States and 1 UT (Odisha has the highest: 13 PVTGs).
33. The **Dhebar Commission (1973)** created the category of Primitive Tribal Groups (PTGs), renamed as PVTGs in 2006.
34. According to Census 2011 religious data, **Hindus** constitute **79.80% (96.63 Crore)** of India's population.
35. **Muslims** constitute **14.23% (17.22 Crore)** of India's population, with the highest decadal growth (**24.6%**) among major religions.
36. **Christians** constitute **2.30% (2.78 Crore)** of India's population and have the highest sex ratio (**1,023**) among all major religious communities.
37. **Sikhs** constitute **1.72% (2.08 Crore)** of India's population and have the lowest sex ratio (**903**) among all major religious communities.
38. **Buddhists** constitute **0.70% (0.84 Crore)** of India's population, with the largest absolute community residing in **Maharashtra (65.3 Lakh)**.
39. **Jains** constitute **0.37% (0.45 Crore)** of India's population and possess the **highest literacy rate (94.9%)** of any religious community in India.
40. Under the Eighth Schedule to the Constitution of India, **22 languages** are recognized.
41. **Hindi** is the most spoken language in India, spoken by **43.63% (52.83 Crore)** of the total population.

42. **Bengali** is the second most spoken language in India (**8.03%**), followed by **Marathi (6.86%)** and **Telugu (6.70%)**.
43. **Sanskrit** is the least spoken Eighth Schedule language in India, with only **24,821 speakers** recorded in Census 2011.
44. The overall Sex Ratio of **Uttar Pradesh** in Census 2011 was **912** (Child Sex Ratio: **902**).
45. In Uttar Pradesh, **Jaunpur** has the highest sex ratio (**1,024**), while **Gautam Buddha Nagar** has the lowest (**851**).
46. In Uttar Pradesh, **Balrampur** has the highest child sex ratio (**950**), while **Baghpat** has the lowest (**841**).
47. In Uttar Pradesh, **Sitapur** has the highest absolute Scheduled Caste population, while **Kaushambi** has the highest SC percentage (**34.7%**).
48. In Uttar Pradesh, **Sonbhadra** has the highest Scheduled Tribe population (both absolute count and percentage: **20.7%**).
49. The **Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act** was enacted in **1994** to prevent sex-selective abortions and arrest the declining child sex ratio.
50. The **Beti Bachao Beti Padhao (BBBP)** national scheme was launched on **22 January 2015 from Panipat, Haryana**, to address the skewed child sex ratio.

⏱️ Timeline / Chronology

Chronological Sequence

```
1950 ⟶ Constitution (Scheduled Castes) Order, 1950 (initially restricted to Hindu faith)
1956 ⟶ Constitutional Amendment extends Scheduled Caste status to Sikhs
1990 ⟶ Constitutional Amendment extends Scheduled Caste status to Neo-Buddhists
2001 ⟶ Census 2001: First detailed electronic religious & mother tongue tabulation
2011 ⟶ Census 2011: Religious demographic matrix (Hindus 79.8%, Muslims 14.2%, Christians 2.3%, Sikhs 1.7%)
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📊 Quick Reference Grid

Table 1: Population Composition by Religion (Census 2011 Master Matrix)

RELIGIOUS COMMUNITY	POPULATION (CRORES)	PROPORTION OF INDIA'S TOTAL (%)	DECADAL GROWTH RATE 2001–2011 (%)	SEX RATIO (FEMALES / 1000 MALES)	EFFECTIVE LITERACY RATE (%)	TOP STATES / UTS WITH HIGHEST POPULATION CONCENTRATION
Hindus	**96.63 Cr**	**79.80%**	**16.8%**	**939**	**73.3%**	Himachal Pradesh (95.2%), Odisha (93.6%), Chhattisgarh (93.2%), Madhya Pradesh (90.9%).
Muslims	**17.22 Cr**	**14.23%**	**24.6%**	**951**	**68.5%**	Lakshadweep (96.6%), Jammu & Kashmir (68.3%), Assam (34.2%), West Bengal (27.0%), Uttar Pradesh (19.3%).
Christians	**2.78 Cr**	**2.30%**	**15.5%**	**1,023** (Highest among major religions)	**84.5%**	Nagaland (87.9%), Mizoram (87.2%), Meghalaya (74.6%), Manipur (41.3%), Goa (25.1%), Kerala (18.4%).
Sikhs	**2.08 Cr**	**1.72%**	**8.4%**	**903** (Lowest among major religions)	**75.4%**	Punjab (57.7%), Chandigarh (13.1%), Haryana (4.9%), Delhi (3.4%), Uttarakhand (2.3%).
Buddhists	**0.84 Cr**	**0.70%**	**6.1%**	**965**	**81.3%**	Sikkim (27.4%), Arunachal Pradesh (11.8%), Mizoram (8.5%), Maharashtra (5.8% — largest absolute count: 65.3 Lakh).
Jains	**0.45 Cr**	**0.37%**	**5.4%**	**954**	**94.9% (Highest Literacy in India)**	Maharashtra (14.0 Lakh), Rajasthan (6.2 Lakh), Gujarat (5.8 Lakh), Madhya Pradesh (5.7 Lakh), Delhi (1.7 Lakh).
Other Religions & Unstated	**1.08 Cr**	**0.90%**	—	—	—	Sarna (Jharkhand/Odisha), Sanamahism (Manipur), Parsis / Zoroastrians (approx. 57,264).

Table 2: Scheduled Castes (SC) vs Scheduled Tribes (ST) Master Demographic Matrix

DEMOGRAPHIC PARAMETER	SCHEDULED CASTES (SC — ARTICLE 341)	SCHEDULED TRIBES (ST — ARTICLE 342)	KEY EXAMINATION DISTINCTIONS & TRAPS
Constitutional Provision	**Article 341** (Presidential Order notifies castes).	**Article 342** (Presidential Order notifies tribes).	Both are notified state-wise; there is no national central list without state specification.
Total All-India Population	**20.14 Crore (16.63% of India)**	**10.43 Crore (8.61% of India)**	SC population is approximately double the ST population.
Sex Ratio (2011)	**945** (Higher than national average 943)	**990** (Significantly higher than national average)	Tribal societies exhibit much higher gender parity; **Kerala ST sex ratio is 1,025**; **Arunachal ST sex ratio is 1,032**.
Child Sex Ratio 0–6 Yrs (2011)	**933** (Higher than national average 919)	**957** (Substantially higher than national average)	Child sex ratio in STs is the highest among all social groups in India.
Effective Literacy Rate (2011)	**66.1%** (Male: 75.2%, Female: 56.5%)	**59.0%** (Male: 68.5%, Female: 49.4%)	ST literacy is the lowest among major demographic groups.
State with Highest Absolute Count	**1. Uttar Pradesh (4.14 Crore)** 2. West Bengal (2.15 Crore) 3. Bihar (1.66 Crore)	**1. Madhya Pradesh (1.53 Crore)** 2. Maharashtra (1.05 Crore) 3. Odisha (0.96 Crore)	UP has highest SC population in numbers; MP has highest ST population in numbers.
State/UT with Highest Percentage (%)	**1. Punjab (31.9%)** 2. Himachal Pradesh (25.2%) 3. West Bengal (23.5%)	**1. Lakshadweep (94.8%)** 2. Mizoram (94.4%) 3. Nagaland (86.5%)	Punjab has highest SC concentration; Lakshadweep/Mizoram have highest ST concentration.
States / UTs with NO Population	**Nagaland, Mizoram, Meghalaya, Lakshadweep, Andaman & Nicobar Islands**	**Punjab, Haryana, Chandigarh, Delhi, Puducherry**	**Crucial Trap**: Punjab & Haryana have NO STs; Nagaland & Mizoram have NO SCs.
Uttar Pradesh Snapshot	Population: **4.14 Crore (20.70% of UP)** • Highest District (Count): **Sitapur** • Highest District (%): **Kaushambi (34.7%)** • Lowest District: **Baghpat (11.4%)**	Population: **11.34 Lakh (0.57% of UP)** • Highest District (Count & %): **Sonbhadra (20.7%)** • Districts with zero/negligible ST: **Baghpat, Ayodhya, Kannauj**	Scheduled Castes form over 1/5th of UP; Scheduled Tribes form only a tiny fraction (0.57%).

Table 3: Top Linguistic Groups in India (Eighth Schedule Language Hierarchy)

RANK	LANGUAGE	NUMBER OF SPEAKERS (CRORES)	PERCENTAGE OF TOTAL POPULATION (%)	LANGUAGE FAMILY CLASSIFICATION	KEY STATES / REGIONS SPOKEN
1st	**Hindi**	**52.83 Cr**	**43.63%**	Indo-Aryan	UP, MP, Bihar, Rajasthan, Haryana, Delhi, HP, Uttarakhand, Jharkhand, Chhattisgarh.
2nd	**Bengali**	**9.72 Cr**	**8.03%**	Indo-Aryan	West Bengal, Tripura, Assam (Barak Valley), Andaman & Nicobar Islands.
3rd	**Marathi**	**8.30 Cr**	**6.86%**	Indo-Aryan	Maharashtra, Goa, Dadra & Nagar Haveli.
4th	**Telugu**	**8.11 Cr**	**6.70%**	Dravidian	Andhra Pradesh, Telangana, Yanam (Puducherry).
5th	**Tamil**	**6.90 Cr**	**5.70%**	Dravidian	Tamil Nadu, Puducherry, Sri Lanka diaspora.
6th	**Gujarati**	**5.55 Cr**	**4.58%**	Indo-Aryan	Gujarat, Daman & Diu, Dadra & Nagar Haveli.
7th	**Urdu**	**5.07 Cr**	**4.19%**	Indo-Aryan	UP, Bihar, Telangana, Maharashtra, Karnataka, J&K.
8th	**Kannada**	**4.37 Cr**	**3.61%**	Dravidian	Karnataka, Goa, Maharashtra border.
9th	**Odia**	**3.75 Cr**	**3.10%**	Indo-Aryan	Odisha, Jharkhand, West Bengal.
10th	**Malayalam**	**3.48 Cr**	**2.88%**	Dravidian	Kerala, Lakshadweep, Mahe (Puducherry).
Least Spoken	**Sanskrit**	**24,821 speakers**	**0.002%**	Indo-Aryan	Declared classical language; primarily scholarly and ritual use.

🧠 Memory Anchors & Mnemonics

- **Religious Order by Population**: **H-M-C-S-B-J** (Hindu 79.8% > Muslim 14.2% > Christian 2.3% > Sikh 1.7% > Buddhist 0.7% > Jain 0.37%)
- **Religious Growth Rate Order (2001–11)**: **M-J-C-H-S-B** (Muslim 24.6% > Jain / Christian > Hindu 16.8% > Sikh 8.4% > Buddhist -6.1%)
- **Highest Literacy among Religions**: **Jains (86.4% overall, 84.7% female)** lead all communities in India.
- **Top 4 Languages of India**: **H-B-M-T** (Hindi 43.63% > Bengali 8.03% > Marathi 6.86% > Telugu 6.70%)
- **Tribal Concentrations**: MP has highest **absolute ST count** (1.53 Cr); Lakshadweep (94.8%) and Mizoram (94.4%) lead in **ST percentage**.

Chapter 04: Census and Urbanisation Topic 4 — Fertility, Mortality, Health and Population Policies

🎯 Consolidated Must-Score Facts

1. India was the **first country in the world** to launch an official National Family Planning Programme in **1952**.
2. **Total Fertility Rate (TFR)** is the average number of children a woman would bear in her lifetime given current age-specific fertility rates.
3. The demographic **Replacement Level TFR** is **2.1 children per woman**.
4. As per the **National Family Health Survey-5 (NFHS-5, 2019–21)**, India's national TFR declined to **2.0**, dropping below replacement level.
5. In NFHS-5, the rural TFR in India stood at **2.1**, while urban TFR was **1.6**.
6. Among Indian States in NFHS-5, **Sikkim** has the lowest TFR (**1.1**), while **Bihar** has the highest TFR (**3.0**).
7. The Total Fertility Rate of **Uttar Pradesh** in NFHS-5 stood at **2.4** (down from 2.7 in NFHS-4 and 3.8 in NFHS-3).
8. **Crude Birth Rate (CBR)** is calculated per **1,000 mid-year total population**.
9. **Crude Death Rate (CDR)** of India in recent SRS estimates is approximately **6.0 deaths per 1,000 population**.
10. **Infant Mortality Rate (IMR)** is defined as the number of deaths of children under 1 year of age per **1,000 live births**.
11. India's national IMR as per SRS 2020 dropped to **28 per 1,000 live births** (Rural: 31, Urban: 19).
12. **Kerala** has the lowest IMR among Indian States at **6 per 1,000 live births**, while **Madhya Pradesh** has the highest at **43**.
13. The Infant Mortality Rate of **Uttar Pradesh** in SRS 2020 was **38 per 1,000 live births**.
14. **Neonatal Mortality Rate (NMR)** measures deaths of infants occurring within the first **28 days of life** per 1,000 live births.
15. **Early Neonatal Mortality** refers to infant deaths within the first **7 days of life**.
16. **Maternal Mortality Ratio (MMR)** is defined as maternal deaths per **100,000 (1 Lakh) live births**.
17. India's national MMR as per SRS 2018–20 dropped to **97 per Lakh live births** (down from 130 in 2014–16).
18. **Kerala** has the lowest MMR in India at **19 per Lakh live births**, while **Assam** has the highest at **195**.
19. The Maternal Mortality Ratio of **Uttar Pradesh** in SRS 2018–20 stood at **167 per Lakh live births**.
20. Target 3.1 of the UN **Sustainable Development Goals (SDG)** aims to reduce global MMR to **less than 70 per 100,000 live births by 2030**.
21. Eight Indian States (Kerala, Maharashtra, Telangana, Andhra Pradesh, Tamil Nadu, Jharkhand, Gujarat, Karnataka) have already achieved the SDG target of MMR < 70.
22. **Life Expectancy at Birth (e_0)** in India is currently estimated at **70.8 years** (Females: 72.2 years, Males: 69.5 years).
23. In India, female life expectancy at birth is approximately **2.7 years higher** than male life expectancy.
24. **Kerala** has the highest life expectancy at birth in India (**75.3 years**), while **Uttar Pradesh and Chhattisgarh** have among the lowest (~65.6 years).
25. **Gross Reproduction Rate (GRR)** measures the average number of **female daughters** born to a woman during her lifetime, ignoring mortality.
26. **Net Reproduction Rate (NRR)** measures the number of surviving daughters who will replace their mothers, accounting for female mortality.
27. **NRR = 1.0** indicates that the population has achieved the exact demographic replacement level.
28. The **Child Marriage Restraint (Amendment) Act, 1978** raised the minimum legal age of marriage from 15 to **18 years for females** and 18 to **21 years for males**.
29. The **Prohibition of Child Marriage Act (PCMA)** was enacted in **2006**, replacing the 1929 Act.
30. The **Jaya Jaitly Task Force (2020)** recommended increasing the legal age of marriage for females from **18 to 21 years**.
31. The **National Population Policy (NPP) 2000** was announced in **February 2000** under Prime Minister Atal Bihari Vajpayee.
32. The long-term objective of NPP 2000 was to achieve **population stabilization by the year 2045**.
33. The **National Commission on Population (NCP)** was constituted in May 2000 and is chaired by the **Prime Minister of India**.
34. The **National Rural Health Mission (NRHM)** was launched on **12 April 2005** by Prime Minister Dr. Manmohan Singh.
35. An **ASHA (Accredited Social Health Activist)** is an honorary female community health volunteer deployed at the rate of **1 per 1,000 population** in rural areas.
36. **Janani Suraksha Yojana (JSY)** was launched in April 2005 under NRHM to promote **institutional deliveries** among BPL and SC/ST pregnant women.
37. Under JSY in Low Performing States (like UP), a rural mother receives a cash assistance of **Rs 1,400**, and the ASHA receives **Rs 600**.
38. **Janani Shishu Suraksha Karyakram (JSSK)** was launched on **1 June 2011** to provide completely free, cashless delivery and newborn care.
39. **Mission Indradhanush** was launched on **25 December 2014 (Good Governance Day)** to achieve full immunization coverage against 12 vaccine-preventable diseases.

40. **Pradhan Mantri Matru Vandana Yojana (PMMVY)** was launched on **1 January 2017** under Section 4 of the National Food Security Act, 2013.
41. Under PMMVY, pregnant women receive a direct cash incentive of **Rs 5,000 in DBT installments** for the first living child (and Rs 6,000 for a second girl child).
42. **Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA)** provides fixed-day free comprehensive antenatal check-ups on the **9th of every month**.
43. As per NFHS-5, the proportion of **Institutional Deliveries in India** rose to **88.6%** (up from 78.9% in NFHS-4 and 38.7% in NFHS-3).
44. Institutional deliveries in **Uttar Pradesh** reached **83.4%** in NFHS-5 (up from 67.8% in NFHS-4).
45. The modern **Contraceptive Prevalence Rate (CPR)** among married women aged 15–49 in India is **67%** in NFHS-5.
46. Female sterilization accounts for the overwhelming majority (**38%**) of all modern contraceptive methods used in India.
47. **Ayushman Bharat - PM-JAY** provides health insurance coverage of **Rs 5 Lakh per family per year** for secondary and tertiary care.
48. In September 2024, the Union Cabinet approved the expansion of Ayushman Bharat PM-JAY to provide **free health coverage to all senior citizens aged 70 and above**, irrespective of income.
49. The **National Health Policy (NHP) 2017** set the target of raising government public health spending to **2.5% of GDP by 2025**.
50. **Sample Registration System (SRS)** under the Registrar General of India publishes the official annual data for CBR, CDR, IMR, TFR, and MMR in India.

⏱ Timeline / Chronology

Chronological Sequence

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1952 ⟶ India becomes 1st country globally to launch an official National Family Planning Programme
1976 ⟶ First National Population Policy (Dr. Karan Singh; target marriage age raised)
1977 ⟶ Post-emergency rechristening of "Family Planning" to "Family Welfare"
2000 ⟶ National Population Policy (NPP 2000) under M.S. Swaminathan Committee (TFR 2.1 by 2010, stable by 2045)
2005 ⟶ National Rural Health Mission (NRHM) launched; Accredited Social Health Activist (ASHA) institutionalised
2017 ⟶ National Health Policy 2017 (Public health spending target: 2.5% of GDP)
```

Table 2: Chronological Evolution of India's Population & Health Policies

POLICY / MISSION	YEAR INITIATED	KEY LEADERSHIP / COMMISSION	CORE OBJECTIVES & MILESTONES	STRATEGIC INTERVENTIONS & KEY OUTCOMES
First National Family Planning Programme	**1952** (1st Five-Year Plan)	Planning Commission (Jawaharlal Nehru)	**India was the first country in the world** to launch an official government-sponsored national family planning programme.	Clinical approach; opened family planning clinics and distributed barrier contraceptives.
National Population Policy (NPP) 1976	**April 1976**	Dr. Karan Singh (Union Health Minister)	Addressed rapid population growth; recommended raising legal age of marriage and monetary incentives for sterilization.	Led to the **Child Marriage Restraint (Amendment) Act, 1978** which raised minimum marriage age to **18 years for females and 21 years for males**.
M. S. Swaminathan Committee Report	**1993–94**	Expert Group headed by Dr. M. S. Swaminathan	Drafted a national population policy shifting focus from coercive targets to reproductive health and gender empowerment.	Foundation for NPP 2000; advocated linking population goals with Panchayati Raj and female education.
National Population Policy (NPP) 2000	**February 2000**	Union Cabinet (Atal Bihari Vajpayee)	1. **Immediate**: Address unmet need for contraception and basic health infrastructure. 2. **Medium-Term**: Bring **TFR to Replacement Level (2.1) by 2010**. 3. **Long-Term**: Achieve **Stable Population by 2045** (subsequently pushed to 2070).	Established the **National Commission on Population (NCP)** chaired by the **Prime Minister**. Target: 100% institutional deliveries and universal immunization.
National Rural Health Mission (NRHM)	**12 April 2005**	Ministry of Health & Family Welfare	Provide accessible, affordable, and quality healthcare to rural populations across 18 high-focus States.	Created the cadre of **ASHA (Accredited Social Health Activist)** — female community health volunteer (1 per 1,000 rural population). Launched **Janani Suraksha Yojana (JSY)** for institutional delivery.
National Health Mission (NHM)	**2013**	Integrates NRHM (2005) and National Urban Health Mission (**NUHM 2013**)	Universal healthcare across rural and urban poor. Focus on RMNCH+A (Reproductive, Maternal, Newborn, Child plus Adolescent Health).	Free drugs and diagnostics service initiative; expansion of emergency ambulance networks (108 / 102).
National Health Policy (NHP) 2017	**March 2017**	MoHFW	Shift from curative care to **wellness and preventive care**; increase public health expenditure to **2.5% of GDP** by 2025.	Target: Increase life expectancy to 70 years by 2025; reduce IMR to 28 by 2019 and U5MR to 23 by 2025; reduce TFR to 2.1.
Ayushman Bharat (PM-JAY & HWCs)	**September 2018**	National Health Authority (NHA) / MoHFW	Two pillars: 1. **Ayushman Arogya Mandirs (1.5 Lakh HWCs)** for Comprehensive Primary Health Care. 2. **PM-JAY**: Cashless health cover of **Rs 5 Lakh per family per year** for secondary/tertiary hospitalisation.	World's largest government-funded health assurance scheme, covering bottom 40% vulnerable population (expanded in 2024 to all citizens aged 70+).

📊 Quick Reference Grid

Table 1: Master Matrix of Demographic Vital Rates & Mathematical Formulas

VITAL RATE / INDICATOR	STANDARD MATHEMATICAL FORMULA	UNITS / DENOMINATOR	NATIONAL BENCHMARK (SRS / NFHS-5)	KEY EXAM CONCEPT & SIGNIFICANCE
Crude Birth Rate (CBR)	$\frac{\text{Total Live Births in a year}}{\text{Mid-Year Total Population}} \times 1,000$	**Per 1,000 Total Population**	**19.5 per 1000** (Rural: 21.1, Urban: 16.1)	Crude measure because denominator includes children, males, and elderly who do not give birth.
General Fertility Rate (GFR)	$\frac{\text{Total Live Births in a year}}{\text{Total Women in Reproductive Age (15–49)}} \times 1,000$	**Per 1,000 Women Aged 15–49**	**68.7 per 1000**	Refines CBR by restricting the denominator strictly to women of childbearing age.
Age-Specific Fertility Rate (ASFR)	$\frac{\text{Live Births to Women in Age Group } x}{\text{Total Women in Age Group } x} \times 1,000$	**Per 1,000 Women in Age Group**	Peak fertility in India occurs in age group **20–24 years**.	Measures fertility behavior across 5-year reproductive age cohorts (15–19, 20–24, 25–29... 45–49).
Total Fertility Rate (TFR)	$5 \times \sum \text{ASFR} / 1,000$	**Average Children per Woman**	**2.0** (NFHS-5: Rural 2.1, Urban 1.6)	Synthetic measure of lifetime completed family size. **Replacement Level TFR is 2.1**.
Gross Reproduction Rate (GRR)	$\text{TFR} \times \left(\frac{\text{Female Births}}{\text{Total Births}}			
ight)$	**Average Daughters per Woman**	$\approx \mathbf{0.95 - 1.0}$	Average number of female children a woman would bear, ignoring female child mortality.	
Net Reproduction Rate (NRR)	$\sum \left[\text{ASFR}_{\text{Female}} \times \text{Female Survival Probability } (L_x)			
ight]$	**Surviving Daughters per Woman**	**NRR = 1.0** represents exact demographic replacement.	Measures the extent to which a generation of newborn daughters replaces their mothers, accounting for mortality.	
Crude Death Rate (CDR)	$\frac{\text{Total Deaths in a year}}{\text{Mid-Year Total Population}} \times 1,000$	**Per 1,000 Total Population**	**6.0 per 1000** (SRS 2020)	Overall mortality rate; sensitive to age composition of population.
Infant Mortality Rate (IMR)	$\frac{\text{Deaths of Infants under 1 Year of age}}{\text{Total Live Births in that year}} \times 1,000$	**Per 1,000 Live Births**	**28 per 1000** (Rural: 31, Urban: 19)	Premier indicator of socio-economic development, maternal nutrition, and primary healthcare quality.
Under-Five Mortality Rate (U5MR)	$\frac{\text{Deaths of Children under 5 Years of age}}{\text{Total Live Births in that year}} \times 1,000$	**Per 1,000 Live Births**	**32 per 1000** (SRS) / 41.9 (NFHS-5)	Probability of dying before reaching exactly five years of age.
Maternal Mortality Ratio (MMR)	$\frac{\text{Maternal Deaths from pregnancy-related causes}}{\text{Total Live Births in that year}} \times \mathbf{100,000}$	**Per 100,000 Live Births (Lakh)**	**97 per Lakh Live Births** (SRS 2018–20)	**CRITICAL EXAM TRAP**: Denominator is **100,000 Live Births**, NOT 1,000! SDG 3.1 target is MMR < 70.
Life Expectancy at Birth (e_0)	Derived from demographic Life Tables	**Years of Life Expected**	**70.8 Years** (Females: 72.2 yrs, Males: 69.5 yrs)	Average number of years a newborn infant would live if prevailing mortality patterns at birth remain constant.

Table 3: State-Wise Health & Demographic Disparities in India

DEMOGRAPHIC METRIC	TOP PERFORMING STATES (BEST)	BOTTOM PERFORMING STATES (WORST / HIGH BURDEN)	UTTAR PRADESH POSITION
Total Fertility Rate (TFR — NFHS-5)	**1. Sikkim (1.1)** 2. Goa & Ladakh (1.3) 3. West Bengal & Kerala (1.5) 4. Tamil Nadu & Punjab (1.6)	**1. Bihar (3.0)** 2. Meghalaya (2.9) 3. Uttar Pradesh (2.4) 4. Jharkhand (2.3) 5. Manipur (2.2)	**TFR: 2.4** (Down from 2.7 in NFHS-4 and 3.8 in NFHS-3). Still above replacement level (2.1).
Infant Mortality Rate (IMR per 1000)	**1. Kerala (6)** 2. Mizoram (12) 3. Goa (13) 4. Tamil Nadu (13)	**1. Madhya Pradesh (43)** 2. Uttar Pradesh (38) 3. Assam (36) 4. Chhattisgarh (35) 5. Odisha (34)	**IMR: 38 per 1000** (Substantially improved from 67 in 2008, but 2nd highest in India).
Maternal Mortality Ratio (MMR per 1 Lakh)	**1. Kerala (19)** 2. Maharashtra (33) 3. Telangana (43) 4. Andhra Pradesh (45) 5. Tamil Nadu (54)	**1. Assam (195)** 2. Madhya Pradesh (173) 3. Uttar Pradesh (167) 4. Chhattisgarh (137) 5. Odisha (119)	**MMR: 167 per Lakh Live Births** (Down from 359 in 2007–09; high-priority intervention state).
Institutional Deliveries (%)	**1. Kerala (99.8%)** 2. Tamil Nadu (99.5%) 3. Goa (99.5%)	**1. Nagaland (45.7%)** 2. Meghalaya (58.1%) 3. Bihar (76.2%)	**Institutional Deliveries: 83.4%** in NFHS-5 (up from 67.8% in NFHS-4 and 20.6% in NFHS-3).

🧠 Memory Anchors & Mnemonics

- **Vital Rate Denominators**:
- **CBR & CDR**: Total mid-year population ($ imes 1,000$)
- **IMR & MMRate**: Number of **live births** ($ imes 1,000$)
- **MMRatio (Examiner Trap!)**: Number of live births ($ imes 100,000$ or $1 ext{ Lakh}$ live births)
- **GFR**: Women of reproductive age (15–49 years) ($ imes 1,000$)
- **Replacement Level Fertility**: $TFR = 2.1$ children per woman (achieved when NRR = 1.0)
- **Top / Bottom States in Health**:
- Lowest IMR = **Kerala (6 per 1,000)** | Highest IMR = **Madhya Pradesh (43)**
- National TFR = **2.0** (NFHS-5) | UP TFR = **2.35** (declined from 2.74 in NFHS-4).

Chapter 05: Census and Urbanisation Topic 5 — Migration and Population Distribution

🎯 Consolidated Must-Score Facts

1. As per Census 2011, **45.36 Crore people (37.5% of India's population)** are internal migrants by place of last residence.
2. **Rural to Rural (R $ o$ R)** is the **largest internal migration stream** in India, accounting for over 53% of all internal migrants.
3. The Rural to Rural migration stream is **overwhelmingly dominated by females** due to marriage.
4. **Marriage** is the single largest reason for female migration in India, accounting for **66.7% (two-thirds)** of all female migrants.
5. **Work / Employment** is the single largest reason for male internal migration in India, accounting for approximately **38%** of male migrants.
6. The **Rural to Urban (R $ o$ U)** migration stream in India is **predominantly male-dominated**.
7. **Urban to Rural (U $ o$ R)** is the smallest migration stream in India, representing return migration post-retirement or economic crisis.
8. **Intra-district migration** accounts for the largest share (**62%**) of internal migration in India.
9. **Inter-state migration** constitutes approximately **12%** (5.4 Crore migrants) of total internal migrants in India.
10. **Uttar Pradesh** is the **largest net source State** of out-migrants in India, followed by **Bihar**.
11. **Maharashtra** is the **largest destination State** for inter-state in-migrants in India, followed by the **NCT of Delhi**.
12. **E. G. Ravenstein** published the foundational **"Laws of Migration"** in **1885 / 1889**.
13. According to Ravenstein's laws, **females predominate in short-distance migration**, while **males dominate in long-distance migration**.
14. The **Push-Pull Theory of Migration** was propounded by American demographer **Everett S. Lee in 1966**.
15. **Push Factors** are adverse conditions that compel people to leave their place of origin (unemployment, poverty, drought, crop failure).
16. **Pull Factors** are attractive conditions that draw migrants to a destination (better employment, higher wages, modern healthcare, urban amenities).
17. The **Mobility Transition Model** connecting spatial mobility with the Demographic Transition Model was formulated by **Wilbur Zelinsky in 1971**.
18. The **Todaro Migration Model** (Michael P. Todaro, 1969) demonstrates that rural-urban migration is driven by **expected urban-rural income differentials**.
19. **Arithmetic Population Density** is calculated as: **Total Population divided by Total Geographical Land Area**.
20. India's arithmetic population density in Census 2011 was **382 persons per square kilometer**.
21. The population density of **Uttar Pradesh** in Census 2011 was **829 persons per square kilometer**.
22. **Physiological / Nutritional Density** is calculated as: **Total Population divided by Net Cultivated / Arable Land Area**.
23. Physiological density is a much more accurate measure of population pressure on agricultural food-producing capacity than arithmetic density.
24. **Agricultural Density** is calculated as: **Total Agricultural Population divided by Net Cultivated Land Area**.
25. High agricultural density in developing countries is a direct indicator of **disguised unemployment** in the primary sector.
26. **Bihar** has the highest arithmetic population density among Indian States (**1,106 persons / sq km**).
27. **West Bengal** ranks second in population density among States (**1,028 persons / sq km**).
28. **Arunachal Pradesh** has the lowest population density in India (**17 persons / sq km**).
29. Among Union Territories, **Delhi** has the highest population density (**11,320 persons / sq km**).
30. Among Union Territories, **Andaman & Nicobar Islands** has the lowest population density (**46 persons / sq km**).
31. In Uttar Pradesh, **Ghaziabad** has the highest population density (**3,971 persons / sq km**).
32. In Uttar Pradesh, **Lalitpur** has the lowest population density (**242 persons / sq km**).
33. In Uttarakhand, **Haridwar** has the highest density (**801 / km^2**), while **Uttarkashi** has the lowest (**41 / km^2**).
34. The term **"Ghost Villages"** (*Bhootia Gaon*) in Uttarakhand refers to hill villages completely depopulated due to acute out-migration.
35. The two hill districts in Uttarakhand that recorded **negative decadal population growth** in Census 2011 are **Almora (-1.28%)** and **Pauri Garhwal (-1.41%)**.
36. The **Uttarakhand Rural Development and Migration Commission** was established in **2017** with headquarters at **Pauri** to study reverse migration.
37. **Step Migration** refers to a spatial process where a migrant moves in stages (e.g. from a small village to a local town, then to a regional city, and finally to a metropolis).
38. **Circular / Seasonal Migration** refers to temporary, recurring movements of rural labourers who travel for seasonal farm or construction work and return home.

39. The **Inter-State Migrant Workmen (Regulation of Employment and Conditions of Service) Act** was passed in **1979** (now subsumed under the OSH Code, 2020).
40. The **e-Shram Portal** was launched in **August 2021** by the Ministry of Labour and Employment to register unorganized and migrant workers.
41. The **One Nation One Ration Card (ONORC)** scheme enables nationwide portability of food rations for migrant workers under NFSA, 2013.
42. **Feminization of Agriculture** refers to the increasing proportion of agricultural work performed by rural women due to male out-migration to urban industrial centers.
43. As per Census 2011, approximately **83.3% of total internal migrants in India moved within their home state (intra-state)**.
44. **Brain Drain** refers to the cross-border international migration of highly skilled and educated professionals (doctors, engineers, scientists) from developing to developed countries.
45. India has the **largest diaspora in the world**, with over **18 million non-resident Indians** living abroad.
46. India is the **world's top recipient of inward foreign remittances**, receiving over **$100 Billion annually**.
47. The **Pravasi Bharatiya Divas (PBD)** is celebrated in India every two years on **9 January** to mark Mahatma Gandhi's return from South Africa in 1915.
48. The term **"Gravity Model of Migration"** (adapted from Newton's Law of Gravitation) posits that migration volume between two cities is directly proportional to their populations and inversely proportional to the square of the distance between them.
49. **Ecological / Climate Migrants** are individuals forced to migrate due to environmental degradation, rising sea levels, desertification, or natural disasters.
50. In the Northern Plains of India, the high concentration of population is primarily driven by **flat fertile alluvial terrain, perennial Himalayan rivers, and favourable agricultural climate**.

⏱️ Timeline / Chronology

Chronological Sequence

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1885 ⟶ E.G. Ravenstein publishes "The Laws of Migration" (Distance-decay, step-migration)
1940 ⟶ S.A. Stouffer proposes Theory of Intervening Opportunities
1946 ⟶ George K. Zipf formulates the Gravity Model of Migration (P1 × P2 / D)
1966 ⟶ Everett S. Lee propounds the Push-Pull Theory (Origin, Destination, Intervening Obstacles, Personal factors)
1969 ⟶ Michael Todaro formulates the Rural-Urban Expected Income Migration Model
2011 ⟶ Census 2011: 45.6 Crore internal migrants recorded in India (37.7% of total population)
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📊 Quick Reference Grid

Table 1: Four Streams of Internal Migration in India (Census 2011 Matrix)

MIGRATION STREAM	MAGNITUDE / SHARE IN TOTAL MIGRANTS	GENDER DOMINANCE & DEMOGRAPHIC PROFILE	PRIMARY UNDERLYING REASONS & DRIVERS	KEY GEOGRAPHIC & ECONOMIC CORRIDORS
1. Rural to Rural (R $ o$ R)	**Largest Stream** (~53% to 55% of all internal migration)	**Overwhelmingly Female (approx. 70%–75% female)**	**Marriage Migration (Patrilocal exogamy)** is the primary cause for women. Seasonal agricultural harvesting labour for men.	Intra-district and inter-district movements in agrarian plains (UP, Bihar, MP, Rajasthan, West Bengal).
2. Rural to Urban (R $ o$ U)	**Second Largest Stream** (~20% to 22%)	**Predominantly Male (Male selective)**	**Economic / Employment Search**: Rural distress, agricultural underemployment, higher urban wage differentials, modern education.	UP/Bihar $ o$ Delhi-NCR, Mumbai, Surat, Ahmedabad, Bengaluru, Chennai, Kolkata.
3. Urban to Urban (U $ o$ U)	**Third Stream** (~14% to 16%)	**Balanced Male and Female**	Career progression, administrative transfers, higher specialized education, business expansion, step-migration to megacities.	Tier-2/Tier-3 cities $ o$ Tier-1 Metropolises (e.g., Lucknow/Patna $ o$ Delhi/Noida; Pune/Nagpur $ o$ Mumbai).
4. Urban to Rural (U $ o$ R)	**Smallest Stream** (~5% to 6%)	Balanced	Return migration post-retirement, economic shocks/job loss in cities, reverse distress migration during crises (e.g. COVID-19 lockdowns).	Industrial hubs $ o$ Native rural villages across Eastern UP, Bihar, Jharkhand, Odisha.

Table 2: Mathematical Formulations of Population Density Types

DENSITY TYPE	EXACT MATHEMATICAL FORMULA	NUMERATOR	DENOMINATOR	REAL-WORLD ECONOMIC & ANALYTICAL PURPOSE
Arithmetic Population Density	$\frac{\text{Total Population}}{\text{Total Geographical Area}}$	Total headcount of people in the territory	Total land area (including barren, forest, urban, and arable land) in km^2	Crude, universally reported standard density. (India 2011 = **382 persons / km^2**; UP = **829 / km^2**).
Physiological Density (Nutritional Density)	$\frac{\text{Total Population}}{\text{Net Cultivated / Arable Land Area}}$	Total population of the country	Only the net arable / cultivated agricultural land area	Measures true population pressure on food-producing arable land. Crucial for assessing carrying capacity.
Agricultural Density	$\frac{\text{Total Agricultural Population}}{\text{Net Cultivated / Arable Land Area}}$	Farmers + Cultivators + Agricultural Labourers	Only the net cultivated land area	Measures land-man ratio among farming workers. High agricultural density indicates severe disguised unemployment.
Economic Density	$\frac{\text{Total Population}}{\text{Total Economic Resources / Wealth Index}}$	Total population	Monetary index of total productive capital and natural resource capacity	Reflects standard of living and resource adequacy relative to population.

Table 3: Major Migration Theories & Conceptual Frameworks

MIGRATION MODEL / THEORY	PROPOUNDER / THEORIST	YEAR & PUBLICATION	CENTRAL POSTULATES & CORE LAWS	KEY EXAMINATION TAKEAWAYS
Laws of Migration	**E. G. Ravenstein** (English Cartographer & Demographer)	*The Laws of Migration* (**1885 / 1889**)	1. Most migrants travel **short distances**. 2. Long-distance migrants prefer **large commercial centres**. 3. Migration proceeds **step-by-step** (Village $ o$ Town $ o$ City $ o$ Metropolis). 4. Each main migration current generates a **counter-current**. 5. **Females predominate in short-distance migration**, whereas males dominate in long-distance migration. 6. **Economic motives** are the primary cause of migration.	Foundational empirical framework for spatial demographic mobility studies.
Push-Pull Model of Migration	**Everett S. Lee**	*A Theory of Migration* (**1966**)	Migration is governed by 4 sets of factors: 1. **Factors associated with the area of origin** (Push factors). 2. **Factors associated with the area of destination** (Pull factors). 3. **Intervening Obstacles** (Distance, travel cost, physical barriers, immigration laws). 4. **Personal Factors** (Age, gender, education, risk appetite).	Classifies migration into Push factors (negative forces driving out) and Pull factors (positive attractions drawing in).
Mobility Transition Model	**Wilbur Zelinsky**	*The Hypothesis of the Mobility Transition* (**1971**)	Spatial mobility transitions through sequential stages parallel to the **Demographic Transition Model (DTT)**: • Phase 1 (Pre-modern): Little circulation. • Phase 2 (Early transition): Massive rural-to-urban and international migration. • Phase 3 (Late transition): Urban-to-urban dominates. • Phase 4 (Advanced): Inter-urban and intra-urban circulation dominates.	Connects industrialisation and demographic transition with patterns of human migration.
Todaro Migration Model	**Michael P. Todaro**	*A Model of Labor Migration and Urban Unemployment* (**1969**)	Rural-urban migration is driven by **expected urban-rural income differentials** rather than actual wage differentials, accounting for urban unemployment probability.	Explains why rural migration to cities continues even in the presence of high urban unemployment.

🧠 Memory Anchors & Mnemonics

- **Internal Migration Streams Order**: **R-R > R-U > U-U > U-R** (Rural to Rural leads overwhelmingly due to female marriage migration)

- **Male vs Female Migration Drivers**:
- **Females (65%+)**: Move primarily due to **Marriage** (विवाह).
- **Males (50%+)**: Move primarily for **Work / Employment / Business** (रोजगार/आजीविका).
- **Density Types**:
- **Arithmetic Density**: $ ext{Total Population} / ext{Total Area}$ (India: 382, UP: 829)
- **Physiological Density**: $ ext{Total Population} / ext{Net Cultivated Area}$
- **Agricultural Density**: $ ext{Agricultural Population} / ext{Net Cultivated Area}$.

Chapter 06: Census and Urbanisation Topic 6 — Urbanisation and Urban Development

🎯 Consolidated Must-Score Facts

1. As per Census 2011, India's total **Urban Population** was **37.71 Crore (31.16% of total population)**.
2. The decadal growth rate of the urban population during 2001–2011 was **31.80%** (compared to 12.30% for rural population).
3. **Goa** is the most urbanised State in India, with **62.2%** of its population living in urban areas.
4. **Mizoram** is the second most urbanised State in India (**52.1%**), followed by **Tamil Nadu (48.4%)**.
5. **Himachal Pradesh** is the least urbanised State in India, with only **10.0%** urban population.
6. **Bihar** is the second least urbanised State in India (**11.3%**), followed by Assam (14.1%).
7. Among Union Territories, the **NCT of Delhi** is the most urbanised (**97.5%**), followed by **Chandigarh (97.3%)**.
8. **Maharashtra** has the largest absolute urban population in India (**5.08 Crore**), followed by **Uttar Pradesh (4.45 Crore)**.
9. In Census 2011, there were **4,041 Statutory Towns** and **3,894 Census Towns** in India (total 7,935 towns).
10. The number of **Census Towns** surged by 186% from 1,362 in 2001 to **3,894 in 2011**, driving in-situ rural-to-urban transition.
11. A **Census Town** must have a minimum population of **5,000**, at least **75% male non-agricultural workers**, and a density $\ge \mathbf{400 ext{ persons / km}^2}$.
12. **Class I Towns (Cities)** are defined as urban centres having a population of **100,000 (1 Lakh) and above**.
13. In Census 2011, there were **468 Class I Cities** in India.
14. Class I Cities account for **70.2% of India's total urban population**, demonstrating a top-heavy urban hierarchy.
15. A **Million-Plus Urban Agglomeration / City (Metropolis)** is defined as an urban area with a population of **10 Lakh (1 Million) or more**.
16. In Census 2011, there were **53 Million-Plus Cities / UAs** in India (up from 35 in 2001).
17. **Greater Mumbai Urban Agglomeration** is the most populous urban agglomeration in India with **18.4 Million inhabitants**.
18. **Delhi Urban Agglomeration** is the second most populous UA (**16.3 Million**), followed by **Kolkata UA (14.1 Million)**.
19. A **Megacity** is defined as an urban agglomeration with a population of **10 Million (1 Crore) and above**.
20. In Census 2011, India had **3 Megacities**: Greater Mumbai (18.4M), Delhi (16.3M), and Kolkata (14.1M).
21. The total urban population of **Uttar Pradesh** in Census 2011 was **4.45 Crore (22.27% of UP's total population)**.
22. Uttar Pradesh has **7 Million-Plus Cities** as per Census 2011: (1) Kanpur, (2) Lucknow, (3) Ghaziabad, (4) Agra, (5) Varanasi, (6) Meerut, and (7) Prayagraj.
23. **Kanpur** is the most populous city within municipal limits in Uttar Pradesh.
24. In Uttar Pradesh, **Ghaziabad** is the most urbanised district (**67.6% urban**), followed by **Lucknow (66.2%)**.
25. In Uttar Pradesh, **Shravasti** is the least urbanised district with only **3.5% urban population**, followed by Kushinagar (4.7%).
26. In Uttarakhand, **Dehradun** is the most urbanised district (**55.5% urban**), while **Bageshwar** is the least urbanised (**3.5%**).
27. The total **Slum Population** of India in Census 2011 was **6.55 Crore (17.4% of total urban population)**.
28. **Maharashtra** has the highest slum population in India (**1.18 Crore**), followed by Andhra Pradesh (1.02 Crore).
29. **Dharavi** in Mumbai is one of the largest informal slum settlements in Asia.
30. The **74th Constitutional Amendment Act, 1992** inserted **Part IX-A (Articles 243P to 243ZG)** into the Constitution for Municipalities.
31. The 74th Amendment added the **Twelfth Schedule**, listing **18 functional items** devolved to Urban Local Bodies.
32. The 74th Amendment came into force on **1 June 1993**.
33. Under **Article 243S**, the constitution of **Wards Committees** is mandatory for all municipalities having a population of **3 Lakh or more**.
34. Under **Article 243ZD**, a **District Planning Committee (DPC)** must be constituted in every district (at least **four-fifths / 80%** members elected from Zila Panchayat & Municipalities).
35. Under **Article 243ZE**, a **Metropolitan Planning Committee (MPC)** must be constituted in every metropolitan area with population $\ge 10 ext{ Lakh}$ (at least **two-thirds / 66.7%** members elected).
36. The **Smart Cities Mission** was launched on **25 June 2015** to develop **100 Smart Cities** through Special Purpose Vehicles (SPVs).
37. The **AMRUT Mission (Atal Mission for Rejuvenation and Urban Transformation)** was launched on **25 June 2015** covering **500 cities** with population $> 1 ext{ Lakh}$.
38. **AMRUT 2.0** was launched in **October 2021** with an outlay of **Rs 2.77 Lakh Crore** to provide 100% water security across all statutory towns.
39. **Pradhan Mantri Awas Yojana - Urban (PMAY-U)** was launched on **25 June 2015** to provide pucca houses under four verticals.
40. The **Credit Linked Subsidy Scheme (CLSS)** under PMAY-U provides upfront interest subsidies of 3% to 6.5% on home loans for EWS, LIG, and MIG categories.

41. **Swachh Bharat Mission - Urban (SBM-U)** was launched on **2 October 2014** to eliminate open defecation and achieve scientific waste management.
42. Under Swachh Survekshan, **Indore (Madhya Pradesh)** has been consistently ranked as India's cleanest city for multiple consecutive years.
43. **PM SVANidhi** scheme was launched on **1 June 2020** by MoHUA to provide collateral-free working capital loans of Rs 10,000 to **Urban Street Vendors**.
44. **Urban Sprawl** refers to the rapid, unplanned spatial expansion of a city into its surrounding rural-urban fringe.
45. **Primacy / Primate City Law** was formulated by **Mark Jefferson in 1939**, stating that a country's leading city is disproportionately larger than any other city.
46. The **Rank-Size Rule** of urban settlements was formulated by **George Zipf in 1949** ($P_n = P_1 / n$).
47. **Conurbation** is an extensive continuous urban area formed by the coalescing of several expanding cities and towns, a term coined by **Patrick Geddes in 1915**.
48. **Megalopolis** refers to a massive clustered network of adjacent metropolitan areas, a term coined by **Jean Gottmann in 1961** (e.g. Boston to Washington D.C.).
49. **Gentrification** is the socio-economic process of renovating and improving urban deteriorated neighbourhoods, often displacing lower-income residents.
50. In India, the Ministry of Housing and Urban Affairs (**MoHUA**) is the apex Union ministry responsible for urban development policies, planning, and schemes.

⏱️ Timeline / Chronology

Chronological Sequence

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1901 ⟶ Level of Urbanisation: 10.84% (2.58 Crore urban population)
1951 ⟶ Post-Independence Urbanisation: 17.29% (6.24 Crore urban population)
2001 ⟶ Urbanisation reaches 27.81% (28.61 Crore urban population)
2005 ⟶ Jawaharlal Nehru National Urban Renewal Mission (JNNURM) launched
2011 ⟶ Urbanisation reaches 31.16% (37.71 Crore urban population; 53 Million-Plus Cities)
2015 ⟶ AMRUT, Smart Cities Mission, and PMAY-Urban launched simultaneously
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Table 3: Flagship Central Urban Development Schemes & Policies

NATIONAL URBAN MISSION	LAUNCH DATE & MINISTRY	CORE MANDATE & TARGET BENEFICIARIES	KEY FINANCIAL & STRATEGIC FRAMEWORK
Smart Cities Mission	**25 June 2015** (MoHUA)	Develop **100 Smart Cities** to provide core infrastructure, decent quality of life, sustainable environment, and 'Smart' solutions.	Implemented through dedicated **Special Purpose Vehicles (SPVs)** at city level. Central assistance of Rs 500 Cr per city matched by equal State funding (Total Rs 1,000 Cr per city).
AMRUT (Atal Mission for Rejuvenation and Urban Transformation)	**25 June 2015** (MoHUA)	Ensure universal coverage of **water supply, sewerage/septage networks, storm water drains, urban transport, and green spaces** across **500 cities** (population $> 1 ext{ Lakh}$).	**AMRUT 2.0 (October 2021)**: Outlay of Rs 2.77 Lakh Crore to make all 4,378 statutory towns **100% water secure** and provide 100% sewage coverage in 500 AMRUT cities.
Pradhan Mantri Awas Yojana - Urban (PMAY-U)	**25 June 2015** (MoHUA)	Provide all-weather pucca houses to all eligible urban households ("Housing for All").	Four implementation verticals: 1. **ISSR**: In-situ Slum Redevelopment with private participation. 2. **CLSS**: Credit Linked Subsidy Scheme on home loan interest. 3. **AHP**: Affordable Housing in Partnership with public/private sectors. 4. **BLC**: Beneficiary-led individual house construction.
Swachh Bharat Mission - Urban (SBM-U)	**2 October 2014** (MoHUA)	Eliminate open defecation, achieve 100% scientific municipal solid waste management, and promote behavioral change.	Certified cities as **ODF, ODF+, ODF++, and Water+**. **SBM-U 2.0 (2021)** targets making all Indian cities completely **"Garbage Free"** with Star Ratings.
PM SVANidhi	**1 June 2020** (MoHUA)	Micro-credit facility providing affordable, collateral-free working capital loans to **urban street vendors** affected by COVID-19.	Initial loan of Rs 10,000 (escalating to Rs 20,000 and Rs 50,000 upon timely repayment) with **7% interest subvention** and cashback on digital transactions.

📊 Quick Reference Grid

Table 1: Census Classification of Towns and Size-Classes in India (Census 2011)

TOWN CATEGORY / CLASS	DEMOGRAPHIC CRITERION & POPULATION SIZE	NUMBER OF TOWNS IN INDIA (2011)	SHARE OF INDIA'S TOTAL URBAN POPULATION (%)	KEY EXAM SIGNIFICANCE & CHARACTERISTICS
Statutory Towns (ST)	Defined by legal status: All administrative places having a Municipal Corporation, Municipality, Cantonment Board, or Notified Town Area Committee.	**4,041 Towns** (up from 3,799 in 2001)	Subsumes all legal municipalities	Created by statute/law of State Government; independent of strict demographic ratios.
Census Towns (CT)	Defined by demographic criteria: Must satisfy all 3 conditions simultaneously: 1. Population $\ge \mathbf{5,000}$. 2. $\mathbf{\ge 75\%}$ of male main working population engaged in **non-agricultural** pursuits. 3. Density $\ge \mathbf{400 ext{ persons / km}^2}$.	**3,894 Towns** (Massive surge from 1,362 in 2001)	Represents rapid in-situ rural-to-urban transformation	Administratively still governed as Gram Panchayats despite having urban economic characteristics.
Class I Towns (Cities)	Population $\mathbf{\ge 100,000 ext{ (1 Lakh and above)}}$	**468 Cities** (up from 394 in 2001)	**70.2% of Total Urban Population**	**CRITICAL EXAM FACT**: Over 70% of India's urban population is concentrated in Class I cities (Top-heavy urbanisation).
Class II Towns	Population **50,000 to 99,999**	**474 Towns**	**8.5%**	Secondary trade and agro-processing regional hubs.
Class III Towns	Population **20,000 to 49,999**	**1,373 Towns**	**11.1%**	Tehsil headquarters, block service centres.
Class IV Towns	Population **10,000 to 19,999**	**1,683 Towns**	**6.4%**	Large agricultural market centres.
Class V Towns	Population **5,000 to 9,999**	**1,749 Towns**	**3.6%**	Small transitional settlements.
Class VI Towns	Population $\mathbf{< 5,000}$	**424 Towns**	**0.2%**	Small administrative / hill cantonment enclaves.
Million-Plus Urban Agglomerations (Metropolises)	Population $\mathbf{\ge 10 ext{ Lakh (1 Million and above)}}$	**53 UAs / Cities** (up from 35 in 2001)	**42.6% of Total Urban Population**	Top 3: **Greater Mumbai UA (18.4M)**, **Delhi UA (16.3M)**, **Kolkata UA (14.1M)**. Uttar Pradesh has **7 Million-Plus Cities**.
Megacities	Population $\mathbf{\ge 10 ext{ Million (1 Crore and above)}}$	**3 Megacities** in Census 2011	Over 48 Million people	**Greater Mumbai (18.4M)**, **Delhi (16.3M)**, **Kolkata (14.1M)**. (Bengaluru, Chennai, Hyderabad entered later).

Table 2: Urbanisation Disparities Across Indian States & Union Territories (Census 2011)

CATEGORY / INDICATOR	TOP PERFORMING / MOST URBANISED ENTITIES	BOTTOM PERFORMING / LEAST URBANISED ENTITIES	KEY EXAMINATION TAKEAWAYS
Most Urbanised States (% of Urban Population)	**1. Goa (62.2%)** 2. Mizoram (52.1%) 3. Tamil Nadu (48.4%) 4. Kerala (47.7%) 5. Maharashtra (45.2%)	**1. Himachal Pradesh (10.0%)** 2. Bihar (11.3%) 3. Assam (14.1%) 4. Odisha (16.7%) 5. Meghalaya (20.1%)	Goa is India's most urbanised state; Himachal Pradesh is the least urbanised state.
Most Urbanised Union Territories (%)	**1. NCT of Delhi (97.5%)** 2. Chandigarh (97.3%) 3. Lakshadweep (78.1%) 4. Puducherry (68.3%)	**1. Andaman & Nicobar Islands (37.7%)** 2. Dadra & Nagar Haveli (46.7%)	Delhi and Chandigarh are almost completely urbanised.
Largest Absolute Urban Population (Crores)	**1. Maharashtra (5.08 Crore)** 2. **Uttar Pradesh (4.45 Crore)** 3. Tamil Nadu (3.49 Crore) 4. West Bengal (2.91 Crore) 5. Andhra Pradesh (2.78 Crore)	**1. Sikkim (1.54 Lakh)** 2. Arunachal Pradesh (3.13 Lakh) 3. Mizoram (5.71 Lakh)	Maharashtra has the largest urban population in absolute numbers, followed by Uttar Pradesh.
Uttar Pradesh Snapshot	Total Urban Population: **4.45 Crore (22.27% of UP)**. • Most Urbanised District (%): **Ghaziabad (67.6%)**, Lucknow (66.2%). • Least Urbanised District (%): **Shravasti (3.5%)**, Kushinagar (4.7%).	UP has **7 Million-Plus Cities**: Kanpur Nagar, Lucknow, Ghaziabad, Agra, Varanasi, Meerut, Prayagraj (Allahabad).	UP has the second largest urban population in India in absolute numbers, but ranks low in urbanization percentage (22.27%).

🧠 Memory Anchors & Mnemonics

- **Census Town Criteria (3-in-1 Rule)**: 1. Minimum population of **5,000** 2. At least **75% of male main working population** engaged in non-agricultural pursuits 3. Population density of at least **400 persons / sq km** (1,000 / sq mile)
- **Top 5 Million-Plus Urban Agglomerations (2011)**: **M-D-K-C-B** (Greater Mumbai 18.4M > Delhi 16.3M > Kolkata 14.1M > Chennai 8.6M > Bangalore 8.5M)
- **Most / Least Urbanised States**:
- By Percentage: **Goa (62.2%)** > Mizoram (52.1%) > Tamil Nadu (48.4%) | Lowest: **Himachal Pradesh (10.0%)** < Bihar (11.3%)
- By Absolute Number: **Maharashtra (5.08 Cr)** > Uttar Pradesh (4.45 Cr) > Tamil Nadu (3.49 Cr).

Chapter 07: Census and Urbanisation Topic 7 — World Population and Demographic Miscellaneous

🎯 Consolidated Must-Score Facts

1. Global human population officially surpassed **8 Billion** on **15 November 2022** (designated by the UN as the "Day of 8 Billion").
2. **India** officially surpassed China to become the **most populous country in the world** in **April 2023** according to UN DESA estimates.
3. The top 5 most populous countries in the world are: (1) **India**, (2) **China**, (3) **USA**, (4) **Indonesia**, and (5) **Pakistan**.
4. **Asia** is the most populous continent, containing approximately **60% of the total world population**.
5. **Africa** is the second most populous continent (**~18%**) and possesses the highest population growth rate and highest fertility rate (TFR ~4.2).
6. **World Population Day** is observed globally every year on **11 July**.
7. World Population Day was instituted by the Governing Council of UNDP in **1989** to commemorate the "Day of Five Billion" (**11 July 1987**).
8. **Monaco** is the most densely populated sovereign nation in the world (approx. **26,000 persons / sq km**).
9. **Bangladesh** is the most densely populated major country (population $> 10 ext{ Million}$) in the world (approx. **1,160 persons / sq km**).
10. **Mongolia** is the least densely populated sovereign country in the world (approx. **2.1 persons / sq km**).
11. World population reached its first **1 Billion** milestone in the year **1804**.
12. It took **123 years** for the world population to double from 1 Billion to **2 Billion (1927)**.
13. It took only **11 years** for the world population to grow from 7 Billion (2011) to **8 Billion (2022)**.
14. According to UN *World Population Prospects 2024*, the global human population is projected to peak in the **2080s at approximately 10.3 Billion** and then stabilize/decline.
15. The famous phrase **"Development is the best contraceptive"** was coined by Indian Health Minister **Dr. Karan Singh** at the **1974 Bucharest World Population Conference**.
16. The **International Conference on Population and Development (ICPD)** held in **Cairo in September 1994** marked a historic paradigm shift to human rights, reproductive health, and women's empowerment.
17. The **State of World Population** report is published annually by the **United Nations Population Fund (UNFPA)**.
18. The **World Population Prospects (WPP)** is published biennially by the **UN Population Division (UN DESA)**.
19. The **International Institute for Population Sciences (IIPS)** is located in **Deonar, Mumbai** (established 1956).
20. IIPS Mumbai functions as the national nodal agency for executing the **National Family Health Surveys (NFHS)**.
21. **International Women's Day** is celebrated annually on **8 March**.
22. **World Health Day** is observed globally on **7 April** (commemorating the founding of the World Health Organization in 1948).
23. **World Habitat Day** is celebrated every year on the **First Monday of October** (designated by the UN General Assembly).
24. **World Cities Day** is celebrated annually on **31 October**.
25. **International Youth Day** is observed on **12 August** (National Youth Day in India is on **12 January**, Swami Vivekananda's birthday).
26. **International Day of Older Persons** is celebrated globally on **1 October**.
27. **Universal Health Coverage (UHC) Day** is commemorated annually on **12 December**.
28. **National Consumer Rights Day** is celebrated in India on **24 December**, while **World Consumer Rights Day** is on **15 March**.
29. The **Global Gender Gap Report** is published annually by the **World Economic Forum (WEF)**.
30. The **Human Development Report (HDR)** containing the Human Development Index (HDI) is published annually by the **UNDP**.
31. The world's lowest national Total Fertility Rate (TFR) is recorded in **South Korea** (TFR approx. **0.72** in 2023).
32. The world's highest national Total Fertility Rate (TFR) is recorded in **Niger** (TFR approx. **6.7** children per woman).
33. **Japan** has the highest proportion of elderly citizens (aged 65+) in the world (> 29% of its population).
34. The **Tokyo Metropolitan Area (Japan)** is the world's most populous urban agglomeration with approximately **37 Million inhabitants**.
35. The **Delhi National Capital Region (India)** is the second most populous metropolitan area in the world (~33 Million), followed by Shanghai (~29 Million).
36. **Zero Population Growth (ZPG)** is achieved when the Crude Birth Rate equals the Crude Death Rate in the absence of net migration.
37. The **Demographic Winter** refers to the long-term socio-economic crisis resulting from sustained below-replacement fertility, leading to a shrinking, aging workforce.
38. **Sex Ratio at Birth (SRB)** biologically averages approximately **105 male births per 100 female births** globally (or ~952 females per 1000 males).
39. The **United Nations Population Fund (UNFPA)** was established in **1969** and is headquartered in **New York, USA**.
40. The term **"Ecumene"** refers to the permanently inhabited and settled portion of the Earth's surface.

41. The term **"Non-Ecumene"** refers to uninhabited or very sparsely settled areas of the Earth (polar ice caps, hyper-arid deserts, extreme high altitudes).
42. More than **90% of the world's population** lives in the **Northern Hemisphere**.
43. Approximately **80% of the world's population** lives at elevations **below 500 meters above sea level**.
44. Approximately **60% of the world's population** lives within **400 kilometers of a sea coast**.
45. The **Equator-to-Pole population gradient** shows that the temperate mid-latitude zone (20°N to 40°N) contains the highest concentration of human population on Earth.
46. The **World Health Organization (WHO)** was established on **7 April 1948** and is headquartered in **Geneva, Switzerland**.
47. The **United Nations Human Settlements Programme (UN-Habitat)** is headquartered in **Nairobi, Kenya**.
48. The **United Nations Children's Fund (UNICEF)** is headquartered in **New York, USA** (founded 1946).
49. The **United Nations Development Programme (UNDP)** was formed in **1965** and is headquartered in **New York, USA**.
50. **Sustainable Development Goal 3 (SDG 3)** is dedicated to ensuring **"Good Health and Well-Being for All at All Ages"** by 2030.

⏱ Timeline / Chronology

Chronological Sequence

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1804 ⟶ World reaches 1 Billion (Dawn of Industrial Era)
1927 ⟶ World reaches 2 Billion (Took 123 years to double)
1960 ⟶ World reaches 3 Billion (Took 33 years)
1974 ⟶ World reaches 4 Billion (Bucharest World Population Conference)
1987 ⟶ World reaches 5 Billion ("Day of 5 Billion" – 11th July, genesis of World Population Day)
1999 ⟶ World reaches 6 Billion ("Day of 6 Billion" – 12th October)
2011 ⟶ World reaches 7 Billion ("Day of 7 Billion" – 31st October)
2022 ⟶ World reaches 8 Billion ("Day of 8 Billion" – 15th November; UN WPP 2022)
2023 ⟶ India surpasses China as the World's Most Populous Country (UN DESA, April 2023)
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Table 2: Chronology of Global Population "Billion Milestones" (1804 to 2080 Peak)

POPULATION MILESTONE	YEAR REACHED	TIME TAKEN TO ADD 1 BILLION	HISTORICAL & TECHNOLOGICAL ERA	KEY GLOBAL DRIVERS & DEMOGRAPHIC CONTEXT
1 Billion	**1804**	All of human history (~300,000 years)	Dawn of the Industrial Revolution	Agricultural improvements in Europe, initial urban sanitation.
2 Billion	**1927**	**123 Years** (1804–1927)	Inter-war period	Control of major epidemics, clean piped water, early vaccination.
3 Billion	**1960**	**33 Years** (1927–1960)	Post-WWII era	Global deployment of antibiotics (Penicillin), DDT, and modern primary health systems.
4 Billion	**1974**	**14 Years** (1960–1974)	Green Revolution period	Dramatic mortality reduction across Asia and Latin America.
5 Billion	**11 July 1987**	**13 Years** (1974–1987)	Late 20th Century	**Triggered the creation of World Population Day (11 July)**.
6 Billion	**12 October 1999**	**12 Years** (1987–1999)	Turn of the Millennium	Designated as the "Day of 6 Billion" by the United Nations.
7 Billion	**31 October 2011**	**12 Years** (1999–2011)	21st Century expansion	Designated as the "Day of 7 Billion" by the UN.
8 Billion	**15 November 2022**	**11 Years** (2011–2022)	Contemporary Era	Designated as the **"Day of 8 Billion"** by UN DESA.
9 Billion (Projected)	**~2037**	Projected **15 Years**	Slowdown phase	Global fertility decline taking effect (Global TFR ~2.3 in 2024).
10 Billion (Projected)	**~2058**	Projected **21 Years**	Mature phase	Sub-Saharan Africa will drive > 50% of global growth.
Peak Population	**~2080s (~10.3 Billion)**	Peak & plateau	Global contraction phase	Projected by UN WPP 2024 to peak at ~10.3 Billion and then gradually decline.

Table 3: Major UN World Population Conferences & Epochal Declarations

INTERNATIONAL CONFERENCE	YEAR & HOST CITY	KEY LEADERSHIP / SPONSOR	LANDMARK THEME / DOCTRINAL SHIFT	HISTORICAL SIGNIFICANCE & OUTCOMES
World Population Conference (Rome)	**1954** (Rome, Italy)	United Nations	Scientific exchange of demographic data	First global meeting of demographic experts to analyze post-war population trends.
World Population Conference (Belgrade)	**1965** (Belgrade, Yugoslavia)	United Nations	Fertility analysis and family planning	Focused on fertility determinants in developing countries.
World Population Conference (Bucharest)	**1974** (Bucharest, Romania)	United Nations (136 Governments)	**"Development is the Best Contraceptive"**	First inter-governmental conference. Indian delegation led by **Dr. Karan Singh** propounded that socio-economic development automatically lowers fertility. Adopted **World Population Plan of Action**.
International Conference on Population (Mexico City)	**1984** (Mexico City, Mexico)	United Nations (148 Governments)	Integration of population in development planning	Reaffirmed Bucharest goals; focused on primary healthcare, maternal health, and urban growth.
International Conference on Population and Development (ICPD)	**September 1994** (Cairo, Egypt)	United Nations / UNFPA (179 Governments)	**Paradigm Shift: Human Rights & Reproductive Choice**	**Revolutionary Watershed**: Abandoned coercive demographic targets. Placed **women's empowerment, gender equality, and universal reproductive rights** at the centre of population policies. Adopted the 20-year **ICPD Programme of Action (Cairo Consensus)**.

📊 Quick Reference Grid

Table 1: Top 10 Most Populous Countries in the World (UN DESA / WPP Matrix)

GLOBAL RANK	COUNTRY	ESTIMATED POPULATION (2024 / WPP)	SHARE OF WORLD POPULATION (%)	CONTINENT / REGION	CAPITAL CITY	KEY DEMOGRAPHIC CHARACTERISTIC
1st	**India**	**~1.435 Billion** (143.5 Cr)	**~17.8%**	South Asia	New Delhi	Officially surpassed China in April 2023; TFR at 2.0 (below replacement).
2nd	**China**	**~1.419 Billion** (141.9 Cr)	**~17.6%**	East Asia	Beijing	Population entered natural decline in 2022; rapid aging; sub-replacement TFR (~1.0–1.2).
3rd	**United States of America**	**~340 Million** (34.0 Cr)	**~4.2%**	North America	Washington, D.C.	Most populous developed country; steady population growth maintained via net immigration.
4th	**Indonesia**	**~279 Million** (27.9 Cr)	**~3.5%**	Southeast Asia	Jakarta (shifting to Nusantara)	World's most populous island nation and most populous Muslim-majority country.
5th	**Pakistan**	**~245 Million** (24.5 Cr)	**~3.0%**	South Asia	Islamabad	Highest TFR (~3.4) and growth rate among top 10 Asian nations.
6th	**Nigeria**	**~229 Million** (22.9 Cr)	**~2.8%**	West Africa	Abuja	Most populous country in Africa; projected to surpass the USA by 2050.
7th	**Brazil**	**~216 Million** (21.6 Cr)	**~2.7%**	South America	Brasilia	Most populous country in Latin America; TFR has declined to ~1.6.
8th	**Bangladesh**	**~173 Million** (17.3 Cr)	**~2.1%**	South Asia	Dhaka	Most densely populated major nation (> 10 Million pop) in the world (~1,160 / km^2).
9th	**Russia**	**~144 Million** (14.4 Cr)	**~1.8%**	Eurasia (Europe/Asia)	Moscow	Largest geographical territory; suffering from prolonged natural population decline.
10th	**Ethiopia**	**~129 Million** (12.9 Cr)	**~1.6%**	East Africa (Horn of Africa)	Addis Ababa	Second most populous nation in Africa (surpassed Mexico in 2024).

🧠 Memory Anchors & Mnemonics

- **Top 5 Most Populous Countries (2024)**: **I-C-U-I-P** (India > China > USA > Indonesia > Pakistan)
- **UN World Population Conferences**:
- 1954: Rome | 1965: Belgrade | 1974: Bucharest | 1984: Mexico City | 1994: **Cairo (ICPD — paradigm shift to reproductive rights)**
- **World Population Day**: Observed annually on **11 July** (commemorating Day of 5 Billion in 1987)
- **Continents by Population Share**: **Asia (~60%) > Africa (~18%) > Europe (~9.5%) > Latin America (~8%) > North America (~4.7%) > Oceania (<1%)**.